

Adding a reusable feature channel beside a prediction service you cannot change: an evaluated release problem, one confirmed effect, and six negative method results

Integrated revision of 2026-09-19. Branch `research/pcrl-manuscript-integrated-v6`. Study 5 integrated from evidence commit `a56bcc7f`; Study 6 from `ba531ab4`. Its Tier-3 expansion and Tier-4 comparison were never triggered and are reported as such.

Abstract

An organisation already publishes a prediction service—fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We formalise and evaluate what happens when it adds a *second*, reusable feature channel for one recipient without altering a published number, while trying to limit what that recipient, and that recipient colluding with another, can *newly* infer about sex and race. The contribution is an implemented release interface with a measurement contract—per-recipient and coalition views, independently fitted attackers, incremental disclosure always reported beside absolute disclosure—and what five studies measured through it. (i) Exact preservation of the published outputs is structural and holds bitwise on every released view; it does not imply stable service accuracy. (ii) On a locked, previously unused survey year, a penalty built from the coalition’s joint view beats an equal-strength and an equal-total-mass local control under simultaneous 95% intervals in 14 of 16 sensitive cells, none worse; its residence cost is bounded by a point rule only. (iii) Four successive development studies failed to make the mechanism competitive with the strongest channel J : a nonlinear penalty, an exact rotation-invariant repair that disclosed significantly *more*, direct adversarial refinement from a weaker channel, and—in the fifth study, new here— J -initialised adversarial training and a coalition-conditioned rank-reducing projection of J . **No prospectively selected release in the fifth study met its declared competitive conjunction.** Its selection rule returned untouched J for both neural nominees; the projection nominees lowered coalition sex recovery against J at a significant residence cost and did not separate from matched local-only projections. (iv) Adapted closed-form erasers (LEACE, SPLINCE) applied to channels this project already had beat the developed mechanism and are not separated from J . (v) A sixth study inverts the objective—buy *capability* beside a fixed J rather than lower recovery—and its preregistered pilot also fails, informatively: a small appended channel reliably adds capability on a label-free proxy (11–45% of the residual teacher variance) and passes every source allowance, but no configuration holds the added sensitive recovery over J within the declared .001-nat screen; capability and disclosure fall together as the adversarial weight rises. Its expansion tiers were never triggered and are reported as such. We report an empirical and problem-formulation contribution and list every claim of ours this revision withdraws or narrows.

1 What the paper does

Suppose a company already publishes two prediction services: a credit-style model returning probabilities to one business partner, and a coverage model returning probabilities to another. Both partners built systems on those exact numbers, so they cannot be revised. The company now wants to give the first partner a compact reusable feature vector built from what the published predictions leave unexplained, while ensuring that the partner, and the two partners pooling what they hold, cannot use it to infer an applicant’s sex or race better than the published predictions already allow. This paper implements and measures that problem.

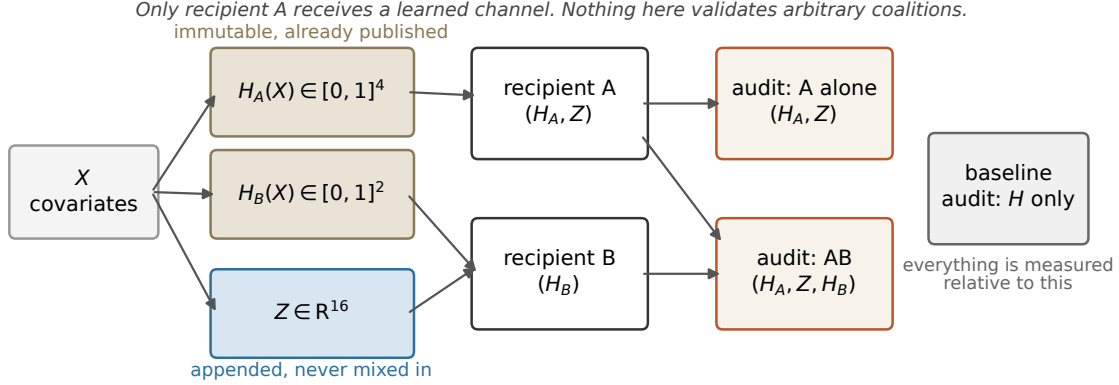


Figure 1: The release graph. H_A and H_B are already published and never modified; Z is appended to A’s wire alone. Three audit views are scored: A alone, the coalition AB, and the H-only baseline every increment is measured against. A schematic, not a result.

- **The interface.** A mechanism appends a channel Z to one recipient’s published output and modifies nothing, so preservation is an identity checked bitwise, not a bounded perturbation.
- **The comparisons it enables.** Heterogeneous mechanisms—spectral moment penalties, adversarially trained neural channels, closed-form erasers, and projections of frozen channels—are compared at matched width on the same wire by one independently fitted attacker slate, with H -only recovery always subtracted and also shown unsubtracted.
- **One confirmed effect.** On a sealed year, a coalition-view penalty lowers measured recovery relative to two matched local-only controls (§6).
- **Six negative method results.** Everything developed afterwards to reach the strongest comparator failed (§7, §8); adapted published erasers beat the developed mechanism. A sixth study, new here, changed objective—buying *capability* beside a fixed J rather than lowering recovery—and its pilot also failed, for a measured reason (§9).

This is an empirical and problem-formulation paper, not a method paper. We do not present a deployable mechanism. We offer no new mathematics: concept erasure, eigenvalue-sign rank selection, minimax adversarial training and conditional-moment characterisations are prior work, and adversarially protected transferable representations are not new [Madras et al., 2018]. The fifth study’s projection is the hard-removal limit of this project’s own residual spectral eigensystem applied to a different input channel—an adaptation of an existing eigensystem, not new eigenvalue mathematics. What we did not find, after a bounded search, is a prior *evaluation* of recipient-specific and combined-access policies over an already-published output that the mechanism may not touch (§4); that is absence of evidence, not a priority claim.

2 The interface, and four distinctions we hold to throughout

Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published first: $H_A(X) \in [0, 1]^4$, the income-over-\$50k and employment probability vectors released to recipient A, and $H_B(X) \in [0, 1]^2$, the coverage vector released to B. A mechanism may add $Z \in \mathbb{R}^{16}$ for A only; A sees (H_A, Z) , B sees H_B , the coalition AB sees (H_A, Z, H_B) . The policy authorises A for income, employment and residence and B for coverage and commute, and forbids sex and race everywhere plus nine cross-role combinations. The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view. A’s inference path uses only A-side inputs; H_B enters fitting and auditing, never inference.

A_0 is an unprotected 16-dimensional neural channel; J is a 16-dimensional adversarially trained channel recovered from earlier work, the strongest comparator in the study, and *internal*. $L1/L2$ are local controls

using only A’s view, matched to the candidate on penalty strength and on total mass, so a coalition effect cannot be confused with “more penalty”; $C1$ adds a penalty from the coalition’s joint view.

Preservation is architectural; accuracy is empirical. Z is concatenated to H_A , never mixed in. That does not keep the service accurate: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between years.

Added capability is measured against the correct H -only view. A channel adds capability only if independently fitted probes do better with (H_A, Z) than with H_A .

Disclosure is a finite-auditor gain, not a bound. It is the improvement a validation-selected attacker achieves over the same family restricted to H : a floor on what is recoverable, never a ceiling.

Recipients have different permitted tasks , so AB’s audit roles are not either singleton’s.

Scope limits. Only recipient A receives a channel, so nothing here is a multi-channel design; and although residence and commute labels are excluded from every representation fit and selection, residence outcomes have been inspected repeatedly across this line of work, so residence is *reserved from training*, not a pristine unseen task.

3 The mechanism family, and what is a theorem here

Let T be a frozen PCA32 teacher. Build $\phi(T)$ from standardised T and 96 random Fourier features [Rahimi and Recht, 2007]; residualise against a degree-2 polynomial basis in the released service probabilities and eigenwhiten to V . Utility is $U = C_{VR}C_{RV}$ for the residualised teacher R ; protection matrices P_j are built from household-grouped cross-fitted nuisances $m_j(H)$ [Chernozhukov et al., 2016] and residuals $\text{onehot}(S_j) - m_j(H)$, aggregated over local roles, coalition roles and a marginal variant. The release is the top-16 eigenvectors of $U_{\text{norm}} - \lambda P$. Studies 2–5 vary this construction (§7, §8).

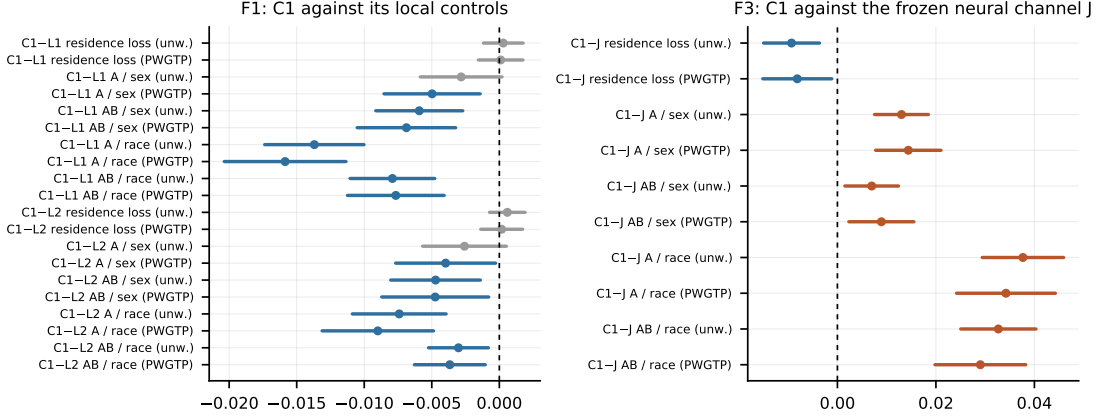
What is proved, and over what. Four elementary statements are proved in Appendix A; nothing else here is a theorem. (1) Appending Z without modifying H preserves H exactly. (2) An unrestricted recipient holding $[H, Z]$ can ignore Z , so its optimal *population* log loss is no worse than with H alone; fitted models need not respect this, and some measured increments are negative. (3) Under Bayes prediction and log loss the population risk difference between H and (H, Z) equals $I(S; Z | H)$; finite validation-selected attackers bound that quantity in neither direction [McAllester and Stratos, 2020]. (4) The Study 3 repaired penalty is exactly invariant to orthogonal reparameterisation (measured to 5.6×10^{-17}). Ky Fan’s principle [Fan, 1949] gives a restricted optimum for the original fixed-matrix family only. The moment condition is a finite fitted subset of the L^2 characterisation of conditional independence used by kernel CI tests [Zhang et al., 2011, Strobl et al., 2019]; vanishing fitted moments establish no conditional independence, and a misspecified $m(H)$ can manufacture a penalty more than ten times the oracle value under *exact* conditional independence.

4 Relation to prior work

SARL [Sadeghi et al., 2019] takes eigenvectors of a utility-minus- λ -dependence matrix; OptNet-ARL [Sadeghi et al., 2021] embeds closed-form players in a learned encoder; K-TOpt [Sadeghi et al., 2022] adds random features and evaluates on Folktables [Ding et al., 2021]; U-FaTE [Dehdashtian et al., 2024] adds a conditional penalty. Our marginal arms are SARL’s construction on whitened residuals, verified by bitwise rebuild. Adversarial fair and transferable representation learning is Madras et al. [2018]. Linear concept erasure [Belrose et al., 2023, Holstege et al., 2025] guarantees erasure against *linear* adversaries on a representation the releaser may edit; adversarial removal leaves recoverable information [Elazar and Goldberg, 2018]; least-privilege learning has fundamental limits [Stadler et al., 2024]. **Reusing a known solver, a conditional-moment characterisation or a rank rule is attribution, not innovation.** Three things differ, with their weight stated honestly: we condition on a released continuous artifact H through cross-fitted residuals rather than stratifying on a label; utility is reconstruction of a residualised teacher; and local and coalition penalties

1: locked 2017 transport <i>confirmed coalition effect</i>	2: nonlinear penalty / rank <i>negative</i>	3: invariant repair + externals <i>negative</i>	4: adversarial from A_0 <i>negative</i>	5: J -init + coalition projection <i>negative</i>	6: utility-first extension of J <i>negative (pilot)</i>
---	--	--	--	--	--

Figure 2: Six studies, kept apart by what each is evidence *for*. Only Study 1 is confirmatory; 2018 and the 2017 partition are development resources for Studies 2–6. ACS 2016 is sealed and unscored. Commits and statuses: Table 4.



paired difference (nats); left of zero favours the candidate. Blue: interval entirely below zero. Orange: entirely above. Grey: crosses zero (unresolved, not unaffected).

Figure 3: Study 1: paired differences with simultaneous 95% intervals inside each pre-declared family, fresh 2017 attacks, both weightings, candidate minus comparator in nats. **Left, F1:** $C1$ against its two local controls; grey A/sex rows are *unresolved*. **Right, F3:** $C1$ against J .

act over a fixed prior output with one recipient receiving a channel. The first two are adaptations. For the third, the nearest candidate, the multi-consumer information-theoretic line [Sankar et al., 2013], designs one sanitised release for many users; we could not confirm a per-recipient release or a collusion term in its text and mark it ambiguous (Appendix B).

5 Evidence status and chronology

California ACS one-year PUMS, ages 19–34, three household splits, disjoint pools for representation fitting, source validation, task fitting and validation, attacker fitting and validation, and evaluation. Attacks are logistic regression, restarted MLPs, gradient-boosted trees and a random-feature kernel family, plus every legal projection of H ’s own attacks into each view. Uncertainty is a paired household-cluster bootstrap with simultaneous intervals inside each pre-declared family. Two weightings (unweighted and person-weighted) are computed from the same predictions and reported separately throughout.

6 Study 1: the confirmed coalition effect, and the rule it did not establish

On the sealed 2017 partition, $C1$ has a simultaneous-interval advantage over both $L1$ and $L2$ under both weightings: of 16 sensitive cells, **14 have adjusted intervals entirely below zero, none above, two unresolved** ($C1$ versus $L1$ on A/sex unweighted, $[-0.00585, +0.00020]$; versus $L2$, $[-0.00568, +0.00052]$). Frozen 2018 attacks reproduce the ordering. **The paired residence rule is a point criterion only:**

Arm	wt.	additional recovery vs. riv16-C1				residence	source
		A / sex	AB / sex	A / race	AB / race	loss diff.	allow.
LEACE	unw.	−0.0081▼	−0.0086▼	−0.0131▼	−0.0158▼	+0.0036	3/3
	PWGTP	−0.0097▼	−0.0093▼	−0.0154▼	−0.0166▼	+0.0040	
SPLINCE	unw.	−0.0088▼	−0.0077▼	−0.0183▼	−0.0165▼	+0.0089▲	3/3
	PWGTP	−0.0100▼	−0.0084	−0.0201▼	−0.0163▼	+0.0091▲	
OptNet-C1	unw.	−0.0074▼	−0.0110▼	−0.0161▼	−0.0145▼	+0.0086▲	2/3
	PWGTP	−0.0089▼	−0.0117▼	−0.0183▼	−0.0151▼	+0.0076▲	
OptNet-L1	unw.	−0.0032	−0.0048	−0.0056	−0.0060	+0.0057	2/3
	PWGTP	−0.0059	−0.0069	−0.0116	−0.0075	+0.0046	
OptNet-L2	unw.	−0.0104▼	−0.0092▼	−0.0184▼	−0.0198▼	+0.0096▲	2/3
	PWGTP	−0.0125▼	−0.0112▼	−0.0210▼	−0.0212▼	+0.0082▲	

Table 1: Study 3: adapted published methods against the repaired mechanism at rank 16 with the coalition policy. Every arm transforms only the auxiliary channel and appends it to unchanged H_A . ▼ adjusted simultaneous interval entirely below zero (the arm recovers *less* than **riv16-C1**); ▲ entirely above. Residence is a *log loss* difference, so a positive value means the arm supplies *less* residence capability. ACS 2018 development pools; three seeds. The last column counts seeds clearing the legacy service-probe allowance, out of 3 (**riv16-C1** clears 1/3).

seed-mean($C1 - \text{comparator}$) ≤ 0.001 passes (values 0.00010–0.00059), but zero of four comparisons support interval noninferiority at that margin, and zero support equivalence. On the two unresolved cells up to 5×10^{-4} nats of extra local sex recovery is not excluded. **And the mechanism is not competitive with J :** $C1$ is significantly worse than J on all four sensitive endpoints under both weightings and better only on residence, $-0.0093 [-0.0150, -0.0036]$. Since J supplies less residence capability, neither Pareto-dominates, and J is internal, so this is evidence about J , not the state of the art.

7 Studies 2–4: three refinements, all negative

Study 2 (nonlinear penalty, rank reduction) moved recovery in the intended direction but failed all eight registered coordination cells; its best candidate stayed significantly worse than J on local sex ($+0.01065$ unweighted, $+0.01210$ person-weighted), and at rank 8 with $L1$ the nonlinear penalty made A/race significantly worse, $+0.00862 [+0.00168, +0.01556]$. After evaluation it was diagnosed that the penalty was *coordinate-dependent*. **Study 3** repaired that exactly (rotation-only share 1.4×10^{-15} over 18 cells, verified before any score was read)—and the repaired releases disclosed *more*: 8 of 48 sensitive cells significantly worse, none better (Appendix E). The repair changed five ingredients at once, so no single cause is isolated.

Study 3 also ran the first external comparisons. LEACE [Belrose et al., 2023] and SPLINCE [Holstege et al., 2025] were applied to the 16-coordinate channel alone—never to the concatenated wire, since editing it would change the published output—so their linear guarantee covers that fitted erasure, not nonlinear prediction from the view an attacker holds; a fixture shows a quadratic feature still recovering after linear $R^2 < 10^{-8}$. Adapted LEACE, OptNet-C1 and OptNet-L2 recover significantly less than the repaired mechanism on all four sensitive endpoints under both weightings, SPLINCE on seven of eight cells (Table 1). LEACE’s residence cost is unresolved against the repaired arm ($+0.0036 [-0.0028, +0.0100]$) and significant against the channel it transformed ($+0.0098 [+0.0036, +0.0159]$), so it dominates nothing. Against J , the external arms are *not separated* at three seeds—which is not equivalence (Appendix F). Protected-class support in a fitting pool determines the realised width of an erased channel (6, 6, 7 across seeds; one race class has counts 1, 1, 0), and an unsupported class is unmeasured, not protected.

Study 4 trained the channel directly against a refreshed attacker ensemble. All 114 trajectories started from A_0 , not J . Across its frontier family 189 cells were significantly worse and 18 significantly better, all 18 on residence utility; its coalition advantage held robustly only against the weak control $L1$ and not under the simultaneous correction against the strength-matched $L2$. 63 of 114 trajectories published their bitwise- A_0 step-0 checkpoint although every one had moved: a selection outcome, not stasis. Its scope correction showed an unmatched attack opportunity crediting an *identical* channel with 0.0155 nats of apparent protection—an evaluation correction, not a gain. Tables and figures are in Appendix H.

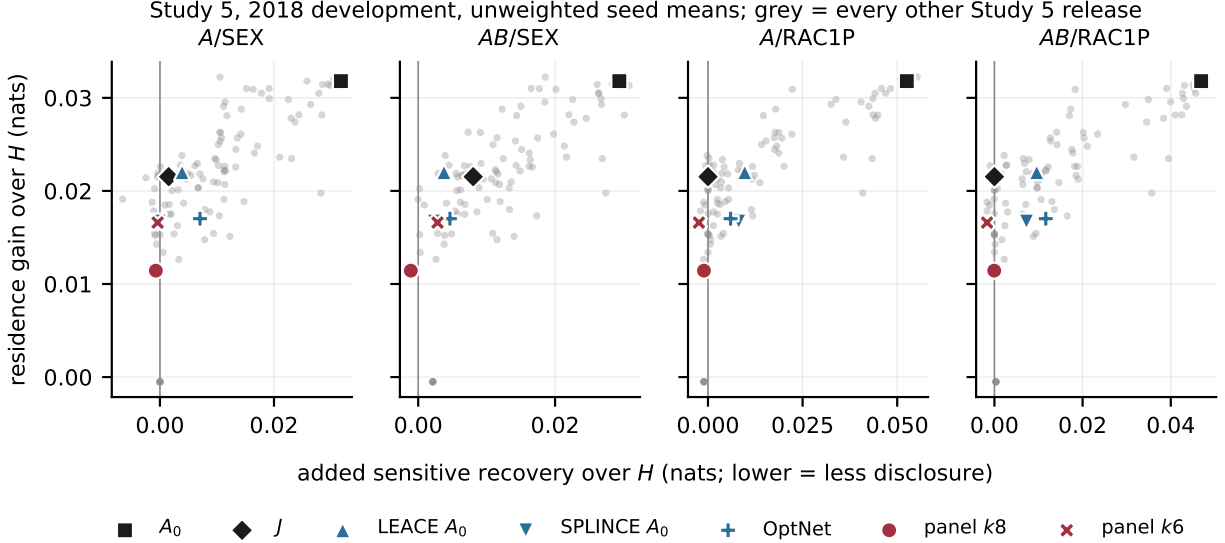


Figure 4: Study 5, 2018 development pools, unweighted seed means: residence gain over H against each sensitive endpoint *separately*—the four risks are never averaged. Red: the two projection nominees; the neural nominees coincide with J . No panel release is up and to the left of J on every panel. Point estimates only; intervals in Figure 8. Person-weighted version: Appendix I.

8 Study 5: J initialisation and coalition-conditioned projection

What was run. Study 5 (evidence commit `a56bcc7f`; protocol locked before any new outcome) asked two questions Study 4 left open. *Track N* (168 nominal slots) is a factorial of adversarial training: initialisation from A_0 or from J ’s actual trained weights (loaded and asserted bitwise identical to J ’s released wire), a common frozen A_0 teacher with coefficient $\gamma \in \{0, .01, .1, 1\}$, penalty $\beta \in \{1, 3\}$ (plus $\beta = 0$ continuations), and policies $L1$, $L2$, $C1$. *Track E* (204 slots) is post-processing of a frozen channel (A_0 or J): whiten, rank directions by a local (L), coalition (C) or feature-count-matched expanded local (LX) cross-moment kernel built from cross-fitted service-only nuisance residuals, and remove the top $k \in \{1, 2, 4, 6, 8\}$ of 16, with marginal, PCA and random-subspace controls and ordinary LEACE/SPLINCE on J . A panel of two configurations per family was nominated on the *validation* split, without residence or commute, and committed before any panel test table. The registered *competitive conjunction* required a sensitive endpoint significantly better than J and than one of `leace_A0`, `spline_A0`, `optnet16_C1`, with residence noninferior at 0.001 (one-sided adjusted bound), every other sensitive endpoint noninferior at a practical 0.001-nat tolerance, and source eligibility, under both weightings, at one candidate-wide correction level (family P, $m = 400$).

Denominators. 372 nominal slots ($168 + 204$) were all fitted; they resolve into 294 distinct audited 2018 releases and 78 exact bitwise duplicates (66 Track N units whose selection returned the untouched start, 12 constant full-span maps), each inheriting the canonical audit. The six references (`ref_A0`, `ref_J` per anchor) are *outside* the 372 and are not double-counted. The stronger attack ran on 54 interfaces (17 releases plus the H -only view, three anchors; six of the 17 are neural arms bitwise equal to J), and 18 Track E units (six releases, three anchors) were transported to 2017. Zero audits were quarantined (Appendix Table 12).

Verdict. No prospectively selected release established the declared competitive conjunction, under either weighting, with J taken either as the exposure-matched `ref_J` or as the historical J audit. We recomputed that verdict from the committed interval ledger rather than copying it.

The neural nominees are untouched J . At $\gamma \in \{0, .01\}$ the registered equal-budget fresh-probe selection returned step 0— J itself—on every anchor of every J -initialised arm: fine-tuning moved the channel, but no moved checkpoint scored better on the monitor fold. At $\gamma \in \{.1, 1\}$ the A_0 teacher pulled J back toward A_0

and the selected channels leak more than J . So **actual J -initialised training was tested here**—which no earlier study did—and under this objective and selection rule it did not improve on J : low- γ selection returned J , and stronger A_0 preservation moved it toward A_0 . Family-X cells where J -initialisation beats A_0 -initialisation (36 of 192) are J 's head start, not an improvement over J .

The projection nominees trade residence for coalition sex protection, as ordinary erasure does. Removing 8 coalition-ranked directions of J (E_J_C_k8) lowers AB/SEX recovery against J by $-0.0091 [-0.0158, -0.0025]$ unweighted (significant under both weightings) and raises residence loss by $+0.0101 [+0.0023, +0.0179]$ unweighted (significant; person-weighted $+0.0085 [-0.0009, +0.0178]$, unresolved): its residence gain over H is 0.0114 against J 's 0.0215. Lower recovery with lower capability is a trade-off, not dominance. Its point estimates are within ± 0.0011 of ordinary LEACE applied to J on every endpoint unweighted (up to 0.0020 person-weighted); **that is point-estimate proximity, not statistical equivalence**—no equivalence test was run and the residence interval is $[-0.0089, +0.0091]$. E_J_C_k6 separates from J on no endpoint. Against its matched local controls L and LX , neither nominee is significantly better on any endpoint (Figure 8): **coalition projection on J did not establish an advantage over matched local projections**. Removing linear directions is not certified conditional privacy: the projection's only exact property is the fixed-offset stationarity statement below, on training rows, within one linear correction family.

Two further findings, both short of a method. Lowering the teacher coefficient from 1 to 0 (A_0 initialisation, $C1$, $\beta = 1$) cut $A/RAC1P$ recovery from $+0.0201$ to $+0.0038$ and $AB/RAC1P$ from $+0.0182$ to $+0.0078$ at a similar residence gain—a coefficient change in an existing objective, with every A_0 -initialised arm still significantly worse than J somewhere. And on A_0 at $k = 2$, coalition projection lowers race recovery against both the local (-0.0202 on $A/RAC1P$) and the expanded-local control (-0.0182), significant under both weightings *in the exploratory family X where it was found*, never nominated, with the residence difference unresolved—**absence of a significant residence difference is not evidence that utility is preserved**—and the release far from competitive ($A/RAC1P$ $+0.0223$ against J 's $+0.0000$). Whether it survives on a channel as strong as J , which has almost no residual race signal, is open. In neural training, coalition conditioning beat $L1$ in 21 of 128 sensitive cells and the strength-matched $L2$ in **0 of 128** (Appendix I).

The stronger attack was weaker. The prespecified MLP[256, 256, 128], frozen before any panel outcome, recovered *less* than the same recipe on the H -only view on both sex endpoints for every interface—on untouched A_0 , -0.092 on A/SEX where the standard audit finds $+0.032$ —and about half the standard audit's race increment on A_0 . It stressed nothing, was not re-tuned, and **supports no claim of stronger-attack survival**.

2017, exploratory. All 18 transported units were scored (2018 maps applied to frozen 2017 channels; only fresh attackers and probes fitted; no intervals). Both projections leak less than J on AB/SEX and $A/RAC1P$ in all three anchors, but the matched local controls show the same pattern, so 2017 shows no coalition-specific signal. E_J_C_k6 is descriptively interesting: unweighted residence gain 0.0187 against J 's 0.0189. Person-weighted it is 0.0159 against 0.0171, a deficit that already exceeds the 0.001 margin at the point estimate, and its per-anchor residence difference changes sign. It is a hypothesis for a fresh prospective evaluation, not a result, and it does not rescue the 2018 verdict.

A protocol change after outcomes, reported as such. The registered identity check required `ref_A0/ref_J` to reproduce the historical A_0/J audits exactly; it **failed** on three cells, because the historical audits carried an extra derived-candidate space no new release had. Amendment 4, declared *after* the 2018 panel intervals were computed, replaced the asymmetric historical J comparison by the exposure-matched `ref_J`. Both versions are reported and the verdict is identical; an unchanged negative verdict does not erase the amendment's timing.

What the projection is, mathematically. With the channel whitened to v , the release is $z_{\text{out}} = \mu + v(I - U_k U_k^\top)S$: the utility-free, infinite-penalty limit of the project's own residual spectral eigenproblem, applied to a frozen neural channel—a limiting form of an existing eigensystem on a different input, not new

eigenvalue mathematics. The rank-deficient case we flagged before the study (whitened projection deletes off-support components of new rows that LEACE’s subtractive form keeps) was resolved by measurement: the supported rank is 16 of 16 in all 204 Track E fits with zero dropped directions, so the whitened and subtractive forms coincide on training and transported rows. The stationarity statement is proved and tested only for a *fixed-offset* multinomial logit with the cross-fitted offset that defines the residual, the same masks, a correction linear in $b(h) \otimes v_{\text{out}}$, and *no free intercept* (whose gradient cross-fitting does not zero); because partial projections leave non-zero training cross-moments, it does not cover the partial nominees at all, and it says nothing about retrained baselines, nonlinear auditors or population quantities (Appendix J).

9 Study 6: the utility-first extension, and a pilot that failed for a measured reason

Two forms of competitiveness. A release beside H can be competitive in two legitimate ways: *lower recovery at comparable utility*, or *higher utility at comparable protection*. Studies 1–5 registered the first. Study 6 (evidence commit `ba531ab4`; protocol frozen before any outcome) registers the second as a *new development objective*. It is not a correction to any earlier numerical result, and no earlier negative becomes a success under it.

Study 6 ran as a preregistered tier ladder with machine-readable gates: infrastructure and archive (pass), evidence reuse and attack calibration (pass), a 42-unit pilot (**fail**), and then an expansion and a nominee comparison that were **never triggered**, because the protocol forbids opening a larger grid after a failed pilot (Table 19). Those two tiers are absent evidence for an unrun design, not missing evidence for a claimed one, and no partial grid was fitted.

The release, and an inclusion fact that bounds what it could ever show. The release is $[H_A, Z_J, R]$ with H_A , H_B and Z_J asserted byte-identical to the untouched J release on every pool, and R a width-2 learned channel using only A-side inference inputs. Because a recipient can ignore R , appending it cannot reduce optimal population utility *or* optimal sensitive recovery (§3, statement 2): a deterministic append is never privacy amplification of J , and any reading of “appending R protected us” is wrong by construction. The target is therefore extra useful capability at a *bounded empirical* disclosure increase. Capability is measured by a proxy that uses no residential label: reduction in reconstruction error of the residual teacher signal Z_{A_0} that a frozen decoder from (H_A, Z_J) leaves unexplained. Cross-fitted out-of-fold residuals are training artifacts only; the deployed map uses the single final decoder, and no out-of-fold target, row identifier, H_B or sensitive label is reachable at inference.

What the reanalysis of the completed studies added. Before fitting anything new, the stored evidence of the 372-slot study was re-scored under the new utility-first criterion (residence improvement $\geq .003$ nats with an adjusted interval below zero, and one-sided adjusted upper bounds $\leq .001$ on each sensitive endpoint, both weightings), re-adjusted for the new search family ($m = 2480$; the earlier $m = 3900$ adjustment was not reused). **0 of 124 configurations pass against J and 0 against `leace_A0`.** 55 of 124 have a residence point estimate better than J under both weightings, but the closest are all A_0 -channel releases whose sensitive upper bounds are an order of magnitude outside the tolerance (for `E_A0_C_k2`, residence $-0.0094 / -0.0100$ against worst sensitive upper $+0.040$). This removes the reading that a near miss was hiding in the earlier evidence; it does not change that study’s verdict or rules.

The pilot, and the measured reason it failed. 42 units (14 configurations $\times$ 3 anchors) were fitted and audited, 41 distinct plus one exact duplicate, with 9 reference audits, zero quarantines and zero probability-integrity failures. **Every configuration passed the capability leg and the historical source allowances; every configuration failed the disclosure screen.** R reliably carries information the frozen J channel does not—11% to 45% of the residual teacher variance—and it carries sensitive information with it at every β tried. Within each policy, capability and the worst sensitive increment fall together, monotonically, as β rises (Figure 5). The closest configuration, `X_r2_L2_b030`, reaches a worst mean increment of $+0.0016$ nats unweighted ($+0.0021$ person-weighted) and misses the .001 screen under both weightings—and it buys that at

Study 6 pilot: capability and disclosure did not separate (β increases to the left)

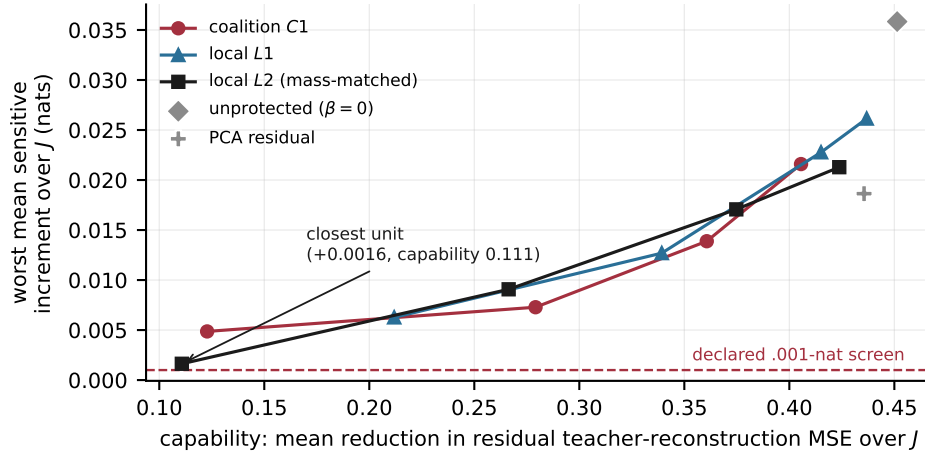


Figure 5: Study 6 pilot, validation split, seed means over three anchors. Capability rises to the right; the worst mean sensitive increment over J across the four endpoints rises upward; the dashed line is the declared .001-nat screen. Increasing the adversarial weight β moves each policy down and to the left—it buys protection by giving back capability—and no configuration reaches the region below the screen with capability left. Point-estimate screening values, not confidence statements.

the lowest capability in the grid (0.111). The unprotected control is the cleanest single row: the best capability (0.451) and the largest disclosure (+0.036). **On this channel, at this width, the extra residence-relevant capability and the extra sensitive recovery were not separable by the adversarial term at any β tried.** That is a statement about this finite family, not a proof that no extension of J can separate them.

No coalition advantage here either. At matched $\beta = 30$ the coalition policy $C1$ is not better than the strength-matched local control $L2$ (worst increment +0.0049 against +0.0016). A third mechanism family now reproduces the pattern that coalition conditioning buys nothing over a strength-matched local control on a strong channel.

A stronger attacker was sought again, and again not found. Eight prespecified attacker recipes were calibrated against H , A_0 , J and LEACE-on- A_0 under a frozen adoption rule; **none beat the standard slate**, so the standard slate was kept and *added stress strength is not established*—the same outcome as Study 5, reached with a different instrument. The positive control passed: the slate recovers A_0 ’s known signal (+0.032 on A/SEX , +0.051 on $A/\text{RAC1P}$ over H).

What a utility-first claim would still require. Increased recovery *bounded* on every declared endpoint by one-sided adjusted intervals—not merely non-significant—with the .001-nat allowance named as an allowance rather than zero harm; comparison against every strong external comparator and local control; and a separation of any utility gain from coalition specificity, since an ordinary non-coalition extension performing equally well is a counterexample to a coalition-specific claim. Study 6 reached none of those levels: nothing was nominated, so no adjusted interval was computed for the pilot, and its thresholds are point estimates. Carrying the frontier further would be an outcome-driven change to a design whose outcome is now known, so it belongs to a new prospective study.

unit	capability	A/SEX	AB/SEX	A/RAC1P	AB/RAC1P	screen
X_r2_C1_b001	0.406	+0.0216	+0.0193	+0.0163	+0.0161	FAIL
X_r2_C1_b003	0.361	+0.0139	+0.0133	+0.0088	+0.0059	FAIL
X_r2_C1_b010	0.279	+0.0073	+0.0057	+0.0026	+0.0041	FAIL
X_r2_C1_b030	0.123	+0.0003	+0.0049	-0.0009	+0.0001	FAIL
X_r2_L1_b001	0.437	+0.0227	+0.0204	+0.0261	+0.0225	FAIL
X_r2_L1_b003	0.415	+0.0228	+0.0206	+0.0118	+0.0114	FAIL
X_r2_L1_b010	0.339	+0.0127	+0.0127	+0.0048	+0.0052	FAIL
X_r2_L1_b030	0.212	+0.0026	+0.0063	-0.0004	+0.0022	FAIL
X_r2_L2_b001	0.424	+0.0213	+0.0193	+0.0174	+0.0178	FAIL
X_r2_L2_b003	0.375	+0.0171	+0.0153	+0.0047	+0.0053	FAIL
X_r2_L2_b010	0.266	+0.0079	+0.0091	+0.0020	+0.0026	FAIL
X_r2_L2_b030	0.111	+0.0016	+0.0016	-0.0000	+0.0007	FAIL
U_r2 unprotected control ($\beta = 0$)	0.451	+0.0247	+0.0225	+0.0358	+0.0301	FAIL
P_r2 deployable PCA residual	0.436	+0.0186	+0.0167	+0.0131	+0.0159	FAIL

Table 2: Study 6 pilot screen, validation split, unweighted seed means of the increment over the same-host re-audit of untouched J ; capability is the mean reduction in residual reconstruction error. Person-weighted values lead to the same decision. The screen is the conjunction of capability, source allowance, mean increment $\leq .001$ on each endpoint and no anchor increment $> .003$. FAIL for all 14.

10 What this evidence does not establish

No attack maximum is an upper bound; a stronger attacker we did not build would move any recovery number upward, and Study 5’s attempt to build one produced a weaker attacker. **Nested validation-selected models need not produce non-negative gains;** we report negative increments unclipped. **No detected difference is not equality:** unresolved cells are never read as parity, and point agreement within 0.0011 is not equivalence. **Ratios are scale-dependent; paired differences are not,** so we quote differences. **Nothing here is a certificate:** removing linear dimensions, or vanishing fitted moments, imply neither $Z \perp S \mid H$ nor any bound on $I(S; Z \mid H)$. **Development against confirmation:** only the 2017 transport of Study 1 is confirmatory; 2018 and the spent 2017 partition are development for Studies 2–5, and **ACS 2016 remains sealed and unscored.** **Sampling and design:** three seeds are three fitted systems on one cohort; bootstraps cover people, not retraining or Census design variance; one state, one age band. **Multiplicity is only as honest as its level:** Study 5 declared one decision level (candidate-wide, family P); its A_0 race effect is reported at the exploratory level where it was found. **A tolerance is not zero harm.** **Numerical integrity:** transient probability-row faults in Studies 3–4 were caught by validation and recomputed; Study 5 quarantined none, but its compaction deleted attacker weights a later stage needed (incident 1), repaired by seeded bitwise-verified regeneration. **Screening thresholds are not confidence statements:** Study 6 nominated nothing, so no adjusted interval was computed for its pilot at all, and its pass/fail legs are point estimates. **Audits do not transport across architectures to better than $\sim .003$ nats:** re-auditing an identical channel on different hardware refits the attacker slate, which moved Study 6’s reference audit by up to 0.0031 nats against its own .003 tolerance. Every Study 6 comparison is therefore made against references re-audited on the same host under the identical slate, and its numbers must not be differenced against earlier studies’ published audit values at a finer precision. **An unrun tier is not evidence:** Study 6’s Tier 3 and Tier 4 were never triggered, and nothing about $r \in \{4, 8\}$, a nominee comparison or a 2017 transport of this mechanism follows from this paper. **A deterministic append cannot protect:** by the inclusion fact, $[H_A, Z_J, R]$ can only weakly increase optimal sensitive recovery over $[H_A, Z_J]$; every measured decrease is a finite-learner artefact.

Every load-bearing claim, its status, the measurement behind it and what it does not establish is in Table 18; the fully traceable list—source commit, artifact, key, pool and uncertainty procedure per claim—is CLAIM_LEDGER.csv.

11 Claims this revision withdraws or narrows

Earlier corrections are carried forward unchanged (Appendix L). The v6 additions, additive to the historical records (CORRECTIONS_V6.md):

1. **“Study 5 has no protocol and no outcome.”** Stale: it is complete and integrated here.
2. **“ J -initialised fine-tuning is untested.”** It was tested in Study 5; under its objective and selection rule it did not improve on J . Earlier studies still did not test it.
3. **The A_0 $k = 2$ race effect “candidate-wide significant”** (the study’s decision summary). It is significant at the exploratory whole-grid level (family X), not in the decision family P, and was not nominated.
4. **“E_J_C_k8 matches LEACE on J within ± 0.0011 .”** Unweighted point estimates only; person-weighted differences reach 0.0020; no equivalence was tested.
5. **“The reference identity check passed.”** It failed on three cells with an established cause; amendment 4 followed the outcomes.
6. **“A sixth, utility-first programme has produced no outcome”** (v5). Stale: its Tiers 0–2 ran and the pilot closed negative; Tiers 3–4 were never triggered.
7. **Study 6’s “best protection achieved was $+0.0016$ nats”.** That is the unweighted worst-endpoint mean; person-weighted the same configuration reaches $+0.0021$. Both miss the .001 screen, so the decision is unchanged, but the headline figure is one weighting.
8. **Audit portability.** Study 6’s cross-platform reproduction of the historical audits exceeded its own .003-nat tolerance (0.0031 on `ref_J`). It is disclosed as a limitation, and its comparisons use same-host references; earlier studies’ published audit values must not be differenced against it more finely.

12 Conclusion

A service that has already published predictions can add a reusable channel beside them without changing a released number, and can steer that channel away from directions a recipient or colluding pair would use to infer sex and race. On a locked year, coalition conditioning beats both of its local controls in 14 of 16 sensitive cells, at a residence cost the data cannot bound below 0.001 nats. Everything built afterwards to make the mechanism competitive with J failed. The fifth study, the most direct test so far, started adversarial training from J itself and projected coalition-conditioned directions out of J : its selection returned J , its projections traded residence for coalition sex protection much as ordinary LEACE on J does, and neither beat matched local-only projections. Reducing teacher strength improved A_0 -based training, and coalition projection showed an exploratory race effect on A_0 ; neither is a competitive method. The sixth study changed the question, asking whether a channel appended beside a fixed J can buy authorised capability at bounded added disclosure. Its pilot answered cleanly for this family: the extension does carry real extra capability, and it carries sensitive information with it at every adversarial weight tried, so the declared operating point was never reached and the expansion was not opened.

The defensible contribution is the release problem and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery side by side, negative increments unclipped, denominators beside every count, per-endpoint results never averaged, and a sealed temporal transport with multiplicity fixed in advance. That contract is what made six negative method results legible rather than ambiguous. The mechanism family, as tested, is not competitive under either definition of competitiveness we have registered. What a sixth study adds is that the failure is not an artefact of asking only for lower recovery: when we asked instead for more capability at bounded disclosure, the same inseparability appeared in a new mechanism, on a proxy that never saw a residential label. Whether a different family can separate them on this channel remains open, and would need a fresh prospective design rather than a larger β on this one.

13 Reproducibility

Deterministic seeds; hash-bound completion records. Study 1’s 90 endpoints were regenerated by an independent scorer to 1.1×10^{-15} ; Studies 3–5 report independent replays at 4.4×10^{-16} (Study 5: 3,780

checks). Every table and figure is generated from committed evidence read through the object store at pinned commits, each digit traceable to a (commit, path, sha256) triple. For v5, every Study 5 number was recomputed from `POINTS_2018.csv`, `INTERVALS_P.csv`, `INTERVALS_X.csv`, `STRESS.json`, `TRACK_E_FITS.json` and `EXPLORATORY_2017.json` at commit 7f961d5c: 113 checks, all agreeing (`STUDY5_VERIFICATION.json` in the v5 review package). For v6, every Study 6 number was re-derived from `PILOT_SCREEN.json`, `TIER1_REANALYSIS.json`, `CALIBRATION.json` and the gate files at commit 0517c06a: 118 checks, all agreeing, including the pilot's pass/fail legs recomputed from the stored per-anchor increments rather than read from its flags (`STUDY6_VERIFICATION.json`). This revision fitted no ACS model.

A Elementary results and their boundaries

Proofs and the exact conditions under which each statement holds are in `MATHEMATICAL_REVIEW.md`, at four levels—population definition, empirical estimator, algorithmic approximation, numerical test—because conflating them is the failure mode this appendix exists to prevent.

B Related-work comparison

Work	fixed prior output	reusable extra features	recipient permissions	coalition-aware training	independent attack eval.
This paper	✓	✓	✓	✓	✓
Madras et al. 2018 (LAFTR)	–	✓	–	–	part.
LEACE [Belrose et al., 2023]	–	–	–	–	–
SPLINCE [Holstege et al., 2025]	–	–	–	–	–
SARL [Sadeghi et al., 2019]	–	–	–	–	part.
OptNet-ARL [Sadeghi et al., 2021]	–	–	–	–	part.
K-TOpt [Sadeghi et al., 2022]	–	–	–	–	part.
U-FaTE [Dehdashtian et al., 2024]	–	–	–	–	✓
Sankar et al. [2013]	–	UNV.	UNV.	UNV.	–
Stadler et al. 2024	–	–	–	–	n/a
Elazar & Goldberg 2018	–	–	–	–	✓

Table 3: The five features this paper’s setting turns on. ✓ the text demonstrates it; – its setting excludes it; “part.” a partial form (an adversarial objective or a transfer test, not an independent attacker slate); UNV. could not be settled from what was read and is **not** guessed from a title or abstract. Several of these methods can be *adapted* into this setting; three were, and they beat the mechanism developed here. Provenance for every row, including which sources were read in full text and which only as abstracts, is in `RELATED_WORK_COMPARISON.md`.

The first column is ours only in the sense that no text we read *evaluated* a mechanism under an immutable published output with per-recipient and coalition audit roles—absence of evidence after a bounded search. Several of these methods were adapted into this setting and beat the mechanism developed here.

C Chronology

study	construction	evidence	status	what it established
1	locked ACS 2017 transport	349efa45	confirmatory	coalition effect over both local controls; the .001 point rule, not interval noninferiority
2	nonlinear penalty / rank	c37807e4	development	negative; penalty coordinate-dependent, diagnosed after evaluation
3	invariant repair + external baselines	73903b7f	development	negative; closed-form erasers beat the developed mechanism
4	direct adversarial refinement from A_0	69e790af	development	negative on both years
5	J -initialised factorial + coalition projection	a56bcc7f	development	negative; no prospectively selected release met the conjunction
6	utility-first residual extension of J	ba531ab4	development	negative pilot: capability gained on the proxy, disclosure screen failed at every configuration; expansion not triggered

Table 4: The six studies, with evidence commits and what each is evidence for.

D Study 1: locked transport, full tables

Contrast	Endpoint	Estimate	Seeds 0 / 1 / 2	Adjusted 95%	
<i>F1 (registered primary): coalition penalty vs local controls, $c = 2.930$ over 20 endpoints</i>					
C1-L1	residence loss	+0.00029	+0.0005 / +0.0004 / -0.0001	[-0.00118, +0.00176]	
C1-L1	A / sex	-0.00282	-0.0046 / -0.0055 / +0.0017	[-0.00585, +0.00020]	
C1-L1	AB / sex	-0.00593	-0.0046 / -0.0094 / -0.0038	[-0.00915, -0.00271]	▼
C1-L1	A / race	-0.01369	-0.0145 / -0.0129 / -0.0137	[-0.01737, -0.01002]	▼
C1-L1	AB / race	-0.00792	-0.0054 / -0.0109 / -0.0075	[-0.01105, -0.00478]	▼
C1-L2	residence loss	+0.00059	+0.0012 / +0.0009 / -0.0004	[-0.00073, +0.00191]	
C1-L2	A / sex	-0.00258	-0.0020 / -0.0051 / -0.0007	[-0.00568, +0.00052]	
C1-L2	AB / sex	-0.00472	-0.0058 / -0.0060 / -0.0024	[-0.00804, -0.00140]	▼
C1-L2	A / race	-0.00741	-0.0054 / -0.0074 / -0.0095	[-0.01088, -0.00394]	▼
C1-L2	AB / race	-0.00302	+0.0004 / -0.0039 / -0.0056	[-0.00524, -0.00081]	▼
<i>F3 (registered secondary): coalition channel vs frozen neural channel J, $c = 2.728$ over 10 endpoints</i>					
C1-J	residence loss	-0.00932	-0.0141 / -0.0020 / -0.0118	[-0.01500, -0.00363]	▼
C1-J	A / sex	+0.01301	+0.0201 / +0.0034 / +0.0155	[+0.00753, +0.01848]	▲
C1-J	AB / sex	+0.00697	+0.0088 / +0.0014 / +0.0107	[+0.00151, +0.01242]	▲
C1-J	A / race	+0.03767	+0.0428 / +0.0383 / +0.0320	[+0.02941, +0.04594]	▲
C1-J	AB / race	+0.03268	+0.0330 / +0.0318 / +0.0332	[+0.02504, +0.04032]	▲

Family	Sens. cells	Better	Worse	Unresolved	advantage fired	c
F1 primary (C1 vs L1, L2; Mode B)	16	14	0	2	4 / 4	2.930
F2 replication (J vs L025, L20; Mode B)	16	6	0	10	4 / 4	3.024
F3 secondary (C1 vs J; Mode B)	8	0	8	0	0 / 2	2.728
F4 frozen transfer (all four; Mode A)	32	21	2	9	6 / 8	3.206

Contrast	Weight	Point est.	Pointwise 95% up.	NI?	Simult. 95% up.	NI?
C1 - L1	unw.	+0.00029	+0.00112	no	+0.00166	no
C1 - L1	PWGTP	+0.00010	+0.00103	no	+0.00164	no
C1 - L2	unw.	+0.00059	+0.00135	no	+0.00182	no
C1 - L2	PWGTP	+0.00017	+0.00102	no	+0.00162	no

Interface	source-probe allowance		half-headroom residence	
	Mode A (frozen)	Mode B (fresh)	Mode A	Mode B
H	3 / 3	3 / 3	0 / 3	0 / 3
E	3 / 3	3 / 3	0 / 3	0 / 3
A0	3 / 3	3 / 3	0 / 3	1 / 3
J	3 / 3	3 / 3	0 / 3	0 / 3
S0	0 / 3	3 / 3	2 / 3	1 / 3
L1	0 / 3	3 / 3	0 / 3	1 / 3
L2	0 / 3	3 / 3	0 / 3	1 / 3
C1	0 / 3	3 / 3	0 / 3	1 / 3

Comparator	gain at $p=1$	p^*	A/race at p^*	reachable?
E	0.0263	1.072	—	no
A0	0.0289	0.976	0.0638	yes
L025	0.0175	1.615	—	no
L20	0.0181	1.555	—	no
J	0.0189	1.493	—	no
S0	0.0304	0.929	0.0688	yes
C1 (target)	0.0282	—	0.0453	—

Interface	Residence gain	A / race		AB / sex	
		absolute	additional	absolute	additional
H	0.0000	0.0165	+0.0000	0.0141	+0.0000
J	0.0189	0.0241	+0.0076	0.0159	+0.0017
E	0.0263	0.0794	+0.0629	0.0299	+0.0158
A0	0.0289	0.0818	+0.0653	0.0393	+0.0251
S0	0.0304	0.0905	+0.0740	0.0397	+0.0256
L1	0.0285	0.0754	+0.0590	0.0288	+0.0146
L2	0.0288	0.0692	+0.0527	0.0275	+0.0134
C1	0.0282	0.0617	+0.0453	0.0228	+0.0087

E Studies 2 and 3: attribution, the repair, erasure rank

Study 2. Both factors moved recovery in the intended direction on different attributes—nonlinear penalty most on sex, rank-8 compression most on race—taking additional A/race recovery from 0.0274 to 0.0055 against J ’s 0.0071 while holding residence gain at 0.0259 against J ’s 0.0215. The penalty summed degree-two monomials $a \leq b$ with equal weight, computing $\sum_a M_{aa}^2 + \sum_{a < b} M_{ab}^2$ rather than $\|M\|_F^2$; only the latter survives $M \mapsto Q^\top M Q$. An information-preserving rotation reaches a mean 0.628 of the refinement’s training-objective gain (range 0.367–0.907), an attained value, not a supremum.

Study 3. The repair used a Frobenius-weighted quadratic moment block and an exact average of three Gaussian kernels on a frozen 512-row subset. Under the repaired objective the repaired map attains a lower value than the defective one in 18 of 18 cells, yet discloses more (Table 5); the largest is +0.0107 [+0.0034, +0.0180] on local race at rank 16. Three of seven registered forecasts were wrong. The repair changed Frobenius weighting, block scaling, centering, the kernel, and the subsets together; surrogate–attacker mismatch remains the live explanation.

Quantity	Defective objective (Study 2)	Repaired objective (Study 3)
Rotation-only share of training gain, attained (<i>a property of the objective</i>)	mean 0.628, range 0.367–0.907 over 18 cells	max. abs. 1.4×10^{-15} over 18 cells
Direct invariance identity $ L(WQ) - L(W) $ on real data	up to 7.5×10^{-3}	5.6×10^{-17}
<i>Measured disclosure, repaired minus defective, over 6 matched contrasts $\times$ 4 sensitive endpoints $\times$ 2 weightings:</i>		
Sensitive cells significantly worse (more recovery)	8 of 48	
Sensitive cells significantly better (less recovery)	0 of 48	
Largest single increase in recovery	+0.0107 A / race, riv16-L2	
Residence cells resolved in either direction	0 of 12	

Table 5: The repair worked on the objective and did not work on the release. The upper rows are a diagnostic of the *objective*; the lower block is a measurement of the *release*. They are reported separately because an exactly invariant objective did not produce less measured disclosure. The attained rotation share is not a supremum – every rotation search stopped on budget exhaustion or a failed line search – and it is not a causal decomposition of what the optimiser did.

Factor	Held at	Sens. cells	Better	Worse	Unres.	Residence better/worse
Nonlinear penalty	$r = 16$	24	7	0	17	0 / 0 (of 6)
Nonlinear penalty	$r = 8$	24	5	1	18	0 / 0 (of 6)
Rank-8 compression	both penalties	48	15	0	33	0 / 0 (of 12)
Coalition vs local penalty	both ranks	64	29	2	33	0 / 0 (of 16)

Endpoint	Weight	Estimate	Adjusted 95%
residence loss	unw.	-0.00435	[-0.01238, +0.00368]
A / sex	unw.	+0.01065	[+0.00331, +0.01799] ▲
AB / sex	unw.	+0.00247	[-0.00396, +0.00890]
A / race	unw.	-0.00160	[-0.01160, +0.00840]
AB / race	unw.	-0.00170	[-0.00832, +0.00492]
residence loss	PWGTP	-0.00432	[-0.01349, +0.00485]
A / sex	PWGTP	+0.01210	[+0.00441, +0.01978] ▲
AB / sex	PWGTP	+0.00422	[-0.00273, +0.01117]
A / race	PWGTP	+0.00133	[-0.00934, +0.01200]
AB / race	PWGTP	+0.00052	[-0.00713, +0.00818]

Seed	q	U : pos / neg / ≈ 0	L1 pos	L2 pos	C1 pos	r_+
0	128	32 / 0 / 96	32	32	32	16
1	128	32 / 0 / 96	32	32	32	16
2	128	32 / 0 / 96	32	32	32	16

Family	Contrast	Endpoint	Weight	Estimate	Adjusted 95%
C1	nlr16-L2-lin16-L2	A / sex	unw.	-0.00597 ▼	[-0.01009, -0.00184]
C1	nlr16-L2-lin16-L2	A / race	unw.	-0.01706 ▼	[-0.02477, -0.00935]
C1	nlr16-L2-lin16-L2	AB / race	unw.	-0.00956 ▼	[-0.01902, -0.00009]
C1	nlr16-L2-lin16-L2	A / race	PWGTP	-0.01316 ▼	[-0.02176, -0.00456]
C1	nlr16-C1-lin16-C1	A / sex	unw.	-0.00639 ▼	[-0.01056, -0.00221]
C1	nlr16-C1-lin16-C1	A / race	unw.	-0.00728 ▼	[-0.01422, -0.00034]
C1	nlr16-C1-lin16-C1	AB / race	unw.	-0.01161 ▼	[-0.02116, -0.00206]
C6	nlr8-L1-lin8-L1	A / race	unw.	+0.00862 ▲	[+0.00168, +0.01556]
C6	nlr8-L2-lin8-L2	A / sex	unw.	-0.00850 ▼	[-0.01237, -0.00463]
C6	nlr8-L2-lin8-L2	A / sex	PWGTP	-0.00634 ▼	[-0.01054, -0.00214]
C6	nlr8-C1-lin8-C1	A / sex	unw.	-0.00653 ▼	[-0.01099, -0.00207]
C6	nlr8-C1-lin8-C1	AB / sex	unw.	-0.00548 ▼	[-0.00858, -0.00238]
C6	nlr8-C1-lin8-C1	AB / sex	PWGTP	-0.00518 ▼	[-0.00876, -0.00159]
C3	lin8-L1-lin16-L1	A / race	unw.	-0.01721 ▼	[-0.02744, -0.00698]
C3	lin8-L1-lin16-L1	AB / race	unw.	-0.01808 ▼	[-0.02775, -0.00842]
C3	lin8-L1-lin16-L1	A / race	PWGTP	-0.01336 ▼	[-0.02493, -0.00178]
C3	lin8-L1-lin16-L1	AB / race	PWGTP	-0.01232 ▼	[-0.02325, -0.00140]
C3	nlr8-L1-nlr16-L1	A / race	unw.	-0.00935 ▼	[-0.01742, -0.00129]
C3	nlr8-L1-nlr16-L1	AB / race	unw.	-0.01133 ▼	[-0.02115, -0.00151]
C3	lin8-L2-lin16-L2	A / race	unw.	-0.02384 ▼	[-0.03547, -0.01221]
C3	lin8-L2-lin16-L2	AB / race	unw.	-0.01359 ▼	[-0.02501, -0.00218]
C3	lin8-L2-lin16-L2	A / race	PWGTP	-0.01811 ▼	[-0.03112, -0.00509]
C3	nlr8-L2-nlr16-L2	A / sex	unw.	-0.00453 ▼	[-0.00903, -0.00002]
C3	lin8-C1-lin16-C1	A / race	unw.	-0.02169 ▼	[-0.03196, -0.01143]
C3	lin8-C1-lin16-C1	AB / race	unw.	-0.01490 ▼	[-0.02559, -0.00421]
C3	lin8-C1-lin16-C1	A / race	PWGTP	-0.01667 ▼	[-0.02864, -0.00470]
C3	nlr8-C1-nlr16-C1	AB / sex	unw.	-0.00636 ▼	[-0.01085, -0.00187]
C3	nlr8-C1-nlr16-C1	A / race	unw.	-0.01463 ▼	[-0.02500, -0.00425]
C4	lin16-C1-lin16-L1	AB / sex	unw.	-0.00515 ▼	[-0.00911, -0.00119]
C4	lin16-C1-lin16-L1	A / race	unw.	-0.01066 ▼	[-0.01642, -0.00489]
C4	lin16-C1-lin16-L1	AB / race	unw.	-0.01427 ▼	[-0.01980, -0.00874]
C4	lin16-C1-lin16-L1	AB / sex	PWGTP	-0.00603 ▼	[-0.01069, -0.00137]
C4	lin16-C1-lin16-L1	A / race	PWGTP	-0.01203 ▼	[-0.01849, -0.00558]
C4	lin16-C1-lin16-L1	AB / race	PWGTP	-0.01502 ▼	[-0.02149, -0.00856]
C4	lin8-C1-lin8-L1	A / race	unw.	-0.01514 ▼	[-0.02051, -0.00976]
C4	lin8-C1-lin8-L1	AB / race	unw.	-0.01108 ▼	[-0.01564, -0.00653]
C4	lin8-C1-lin8-L1	A / race	PWGTP	-0.01535 ▼	[-0.02161, -0.00908]
C4	lin8-C1-lin8-L1	AB / race	PWGTP	-0.00991 ▼	[-0.01535, -0.00446]
C4	nlr16-C1-nlr16-L1	A / sex	unw.	-0.00636 ▼	[-0.01037, -0.00235]
C4	nlr16-C1-nlr16-L1	A / race	unw.	-0.01870 ▼	[-0.02509, -0.01231]
C4	nlr16-C1-nlr16-L1	AB / race	unw.	-0.02253 ▼	[-0.03232, -0.01273]
C4	nlr16-C1-nlr16-L1	A / sex	PWGTP	-0.00728 ▼	[-0.01191, -0.00265]
C4	nlr16-C1-nlr16-L1	A / race	PWGTP	-0.01865 ▼	[-0.02570, -0.01161]
C4	nlr16-C1-nlr16-L1	AB / race	PWGTP	-0.01700 ▼	[-0.02824, -0.00576]
C4	nlr16-C1-nlr16-L2	AB / sex	unw.	+0.00399 ▲	[+0.00034, +0.00764]
C4	nlr8-C1-nlr8-L1	A / sex	unw.	-0.00544 ▼	[-0.00884, -0.00204]
C4	nlr8-C1-nlr8-L1	AB / sex	unw.	-0.00702 ▼	[-0.01050, -0.00353]
C4	nlr8-C1-nlr8-L1	A / race	unw.	-0.02397 ▼	[-0.03285, -0.01510]
C4	nlr8-C1-nlr8-L1	AB / race	unw.	-0.01871 ▼	[-0.02632, -0.01110]
C4	nlr8-C1-nlr8-L1	A / sex	PWGTP	-0.00595 ▼	[-0.00977, -0.00212]
C4	nlr8-C1-nlr8-L1	AB / sex	PWGTP	-0.00645 ▼	[-0.01046, -0.00243]
C4	nlr8-C1-nlr8-L1	A / race	PWGTP	-0.02156 ▼	[-0.03232, -0.01081]
C4	nlr8-C1-nlr8-L1	AB / race	PWGTP	-0.01774 ▼	[-0.02684, -0.00864]
C4	nlr8-C1-nlr8-L2	A / sex	unw.	+0.00229 ▲	[+0.00009, +0.00449]
C4	nlr8-C1-nlr8-L2	AB / sex	unw.	-0.00428 ▼	[-0.00726, -0.00130]
C4	nlr8-C1-nlr8-L2	A / race	unw.	-0.00539 ▼	[-0.01044, -0.00033]
C4	nlr8-C1-nlr8-L2	AB / race	unw.	-0.00868 ▼	[-0.01433, -0.00302]
C4	nlr8-C1-nlr8-L2	AB / sex	PWGTP	-0.00385 ▼	[-0.00734, -0.00037]
C4	nlr8-C1-nlr8-L2	AB / race	PWGTP	-0.01018 ▼	[-0.01717, -0.00319]

Seed	directions deleted	realised width	race class 3 support	cross-cov. after	idempotency	mean pres.
0	10	6	1	4.6×10^{-16}	4.9×10^{-15}	7.7×10^{-15}
1	10	6	1	3.9×10^{-16}	6.7×10^{-15}	4.9×10^{-15}
2	10	7	0	7.2×10^{-16}	3.0×10^{-15}	6.0×10^{-15}

Linear erasure deletes exactly $\text{rank}(\Sigma_{XZ})$ directions. A centred joint one-hot over sex (2), race (9) and coverage (2) has rank $1 + 8 + 1 = 10$, leaving 6 of 16. **Seed 2 differs because one race category has no support in that seed's own fitting pool**, so the centred one-hot has rank 9 and the channel keeps 7 dimensions. Support counts refer to fitting pools, not to the population: a category missing from one pool is not absent from California.

Interface	Objective	r	Fitting labels	Inf. inputs	Role
H	none (no channel)	–	none	H_A	reference
E	adversarial, frozen	16	sex, race	X	context
A0	adversarial, frozen	16	sex, race	X	context
L025	adversarial, frozen	16	sex, race	X	F2 control
L20	adversarial, frozen	16	sex, race	X	F2 control
J	adversarial, frozen	16	sex, race	X	neural ref.
S0	U only ($\lambda = 0$)	16	none	T, H_A	context
M025 / M1	$U - \lambda P_{\text{marg}}$	16	sex, race	T, H_A	context
L1 / L2 / L025	$U - \lambda P_{\text{loc}}$	16	sex, race, cov.	T, H_A	F1/F4 control
C025 / C1	$U - \lambda(P_{\text{loc}} + P_{AB})$	16	+ coal. sex, race	T, H_A	candidate
lin16-*	$U - \lambda P$ (alias)	16	as above	T, H_A	alias, not refitted
nlr16-*	nonlinear refinement	16	as above	T, H_A	dev. only
lin8-*	$U - \lambda P$, compressed	8	as above	T, H_A	dev. only
nlr8-*	nonlinear ref., compressed	8	as above	T, H_A	dev. only

F External comparisons against J

Arm – J	wt.	A / sex	AB / sex	A / race	AB / race	residence
LEACE	unw.	+0.0055	–0.0020	+0.0019	–0.0022	–0.0005
	PWGTP	+0.0056	–0.0016	+0.0034	–0.0027	–0.0010
SPLINCE	unw.	+0.0047	–0.0011	–0.0034	–0.0028	+0.0048
	PWGTP	+0.0052	–0.0007	–0.0013	–0.0025	+0.0041
OptNet-C1	unw.	+0.0061	–0.0044	–0.0012	–0.0009	+0.0045
	PWGTP	+0.0064	–0.0041	+0.0005	–0.0012	+0.0026
OptNet-L1	unw.	+0.0104▲	+0.0017	+0.0093	+0.0077	+0.0017
	PWGTP	+0.0094▲	+0.0008	+0.0073	+0.0064	–0.0004
OptNet-L2	unw.	+0.0031	–0.0026	–0.0035	–0.0062	+0.0056
	PWGTP	+0.0028	–0.0035	–0.0022	–0.0073	+0.0032
riv16-C1	unw.	+0.0135▲	+0.0066	+0.0149▲	+0.0136▲	–0.0041
	PWGTP	+0.0152▲	+0.0077	+0.0188▲	+0.0139▲	–0.0050
riv8-C1	unw.	+0.0121▲	+0.0058	+0.0063	+0.0049	–0.0049
	PWGTP	+0.0147▲	+0.0070▲	+0.0099	+0.0078	–0.0052

Table 6: Study 3 external arms against J . A blank cell is an **unresolved** difference at three seeds, not a demonstrated equality: the adjusted interval contains zero and also contains effects of the size this table reports elsewhere. No equivalence or non-inferiority test was run for these cells. Negative residence means the arm supplies *more* residence capability than J .

All three external point estimates on local sex lean towards J , two missing significance by less than 0.0007; OptNet-L1 is significantly worse than J on local sex. OptNet-ARL’s multi-attribute form is an extrapolation; SPLINCE preserves covariance with the authorised training tasks, an asymmetry in its favour on utility. Shared policy coefficients do not imply equal effective regularisation across methods.

One panel per sensitive endpoint; no score is averaged across them. Filled markers clear the service-probe allowance in 3/3 seeds, hollow do not.

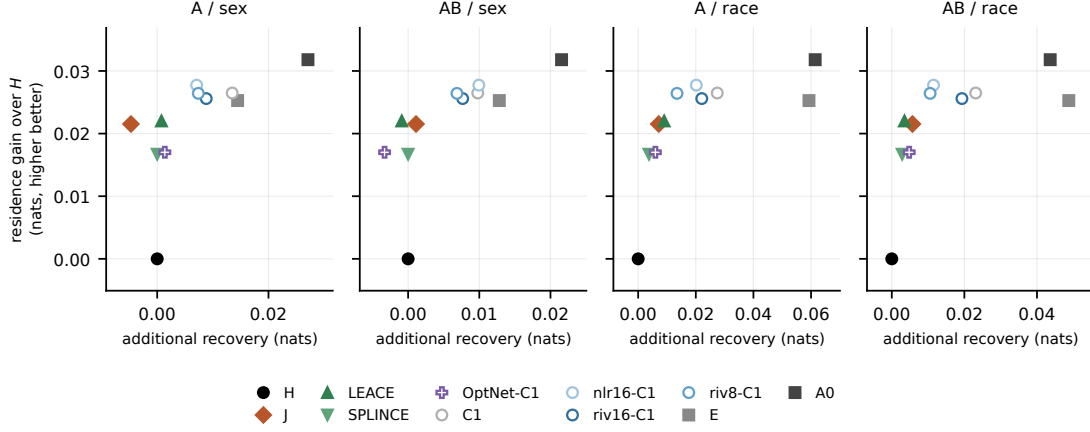


Figure 6: Study 3: authorised capability against additional recovery, one panel per sensitive endpoint.

G Attack scope, and what a paired contrast is invariant to

Condition	fresh independent attacks		+ historical catch-up (reporting scope)	
	absolute	additional	absolute	additional
H	0.0014	+0.0000	0.0071	+0.0000
J	0.0027	+0.0013	0.0024	-0.0047
LEACE	0.0053	+0.0039	0.0078	+0.0008
riv16-C1	0.0159	+0.0145	0.0159	+0.0088
A0	0.0342	+0.0328	0.0342	+0.0271

A / sex, 2018, unweighted, seed means. The reporting scope gives H and the frozen historical interfaces their saved-observer catch-up attacks; the new arms have none, so their *absolute* recovery is unchanged between scopes and only the *subtracted* H baseline moves ($0.0014 \rightarrow 0.0071$). Every paired contrast between two arms is therefore identical on both scales, because the shared baseline cancels; only H -relative levels and ratios are scope-dependent. The absolute recovery of J falls when it is given more attack candidates – a validation-selected maximum is a floor on leakage, never a bound.

H Study 4: full tables

Study 4 used residual-logit attackers on a frozen service-only baseline, an explicit incremental-gain penalty, teacher distortion at coefficient 1.0, $\beta \in \{0.1, 0.3, 1, 3\}$, widths 8 and 16, three policies and three anchors. Against $L2$ its coalition benefit is one endpoint ($AB/RAC1P$) at width 16 at the family-adjusted level only, with residence difference $+0.00346 [+0.00008, +0.00684]$, an interval that crosses the 0.001 reference. Refreshed attackers were kept in 84 of 1350 decisions; training probe and audit agree moderately (Pearson 0.58). 126 nominal fits gave 123 releases, 66 distinct channels and 60 distinct new channels. LEACE at width 8 released a constant channel (measured rank 0).

condition	role	A/SEX	AB/SEX	A/RAC	AB/RAC	residence
A0	no protection	+0.0317	+0.0294	+0.0525	+0.0469	+0.0318
J	frozen neural channel, strongest developed	+0.0015	+0.0080	+0.0000	−0.0006	+0.0215
leace_A0	closed-form eraser on A_0	+0.0039	+0.0038	+0.0097	+0.0096	+0.0220
spline_A0	closed-form eraser on A_0	−0.0002	+0.0024	+0.0081	+0.0073	+0.0167
optnet16_C1	published-method adaptation	+0.0070	+0.0046	+0.0060	+0.0116	+0.0170
spectral_C1	Study 3, previous line	+0.0191	+0.0177	+0.0294	+0.0285	+0.0265
dax8_C1_b300	Study 4, best new arm	+0.0072	+0.0096	+0.0047	+0.0079	+0.0220
dax16_C1_b100	Study 4, coalition width 16	+0.0114	+0.0162	+0.0201	+0.0182	+0.0226
dax16_L2_b300	Study 4, strengthened local	+0.0104	+0.0173	+0.0118	+0.0165	+0.0173
leace_dax8_none	erasure at rank 0	+0.0000	+0.0032	+0.0001	−0.0011	−0.0005

Table 7: Study 4, 2018 development. Seed means over three anchor seeds, matched-exposure scope `kernel_expanded_independent`, audit budget 360, unweighted. Recovery is additional recovery over H ; residence is utility gain over H . Lower recovery is less disclosure, higher residence is more capability. Recomputed cell by cell from `PER_SEED.csv`; all 50 cells agree with the study’s own table to within 5×10^{-5} .

condition	role	A/SEX	AB/SEX	A/RAC	AB/RAC	residence
A0	no protection	+0.0325	+0.0312	+0.0632	+0.0564	+0.0289
J	frozen neural channel, strongest developed	+0.0013	+0.0069	+0.0039	+0.0015	+0.0189
leace_A0	closed-form eraser on A_0	+0.0071	+0.0072	+0.0145	+0.0101	+0.0219
spline_A0	closed-form eraser on A_0	+0.0021	+0.0009	+0.0053	+0.0049	+0.0148
optnet16_C1	published-method adaptation	+0.0093	+0.0088	+0.0211	+0.0185	+0.0139
spectral_C1	Study 3, previous line	+0.0170	+0.0148	+0.0471	+0.0436	+0.0282
dax16_C1_b100	Study 4, coalition width 16	+0.0176	+0.0167	+0.0259	+0.0231	+0.0252
leace_dax8_none	erasure at rank 0	−0.0002	−0.0002	−0.0002	+0.0003	−0.0007

Table 8: Study 4 panel on 2017, exploratory. **Exploratory cross-year development.** The 2017 `final_evaluation` partition is spent; no intervals are computed. Computing them would not be forbidden—they would be conditional descriptive sampling uncertainty—but they would not restore confirmatory status or account for adaptive selection. The original locked 2017 transport result keeps its historical status and is neither restated nor overwritten.

contrast		sensitive endpoints better, of 4		residence difference		
C_1 arm	local control	family-adj.	cand.-wide	estimate	family-adj. interval	cand.-wide
dax16_C1_b100	dax16_L1_b100	4	3	+0.00717	[+0.00305, +0.01129]	worse
dax16_C1_b100	dax16_L2_b100	1	0	+0.00346	[+0.00008, +0.00684]	n.s.
dax8_C1_b100	dax8_L1_b100	4	4	+0.00310	[+0.00041, +0.00579]	n.s.
dax8_C1_b100	dax8_L2_b100	0	0	+0.00002	[−0.00233, +0.00237]	n.s.

Table 9: Study 4 coalition against local conditioning at $\beta = 1$, at both correction levels. Unweighted, $\beta = 1.0$, matched-exposure scope. *Family-adjusted* is the single-step $\max\{-t\}$ interval inside the `coalition_vs_local` family; *candidate-wide* is the simultaneous studentized Bonferroni bound over the full 850-row family, which is the standard the frontier comparison in Table 7 is held to. The study’s own report gives this family at the family-adjusted level only and omits the fourth row, which is the width-8 comparison against the strength-matched control and is null throughout. The coalition benefit is robust against the weak control L_1 and does not survive against L_2 under the candidate-wide standard. L_2 matches C_1 ’s nominal total group weight, not its gradients and not its difficulty.

quantity	count	note
nominal transform fits planned	126	the registry, not a count of systems
released interfaces	123	3 SPLINCE width-8 arms SCOPED INFEASIBLE
bitwise duplicates within a seed	57	preregistered checkpoint rule returned the unmoved channel
distinct channels	66	123 – 57
of those, bitwise identical to a historical arm	6	max abs difference 0.0; reused with proof
distinct and new	60	the number of new systems this study contributes
main arms bitwise identical to no protection	39 of 72	18 at $\beta = 0.1$, 18 at 0.3, 3 at 1.0

Table 10: Study 4, from nominal slots to distinct new systems. Nominal registry slots are not distinct models. The duplicates are outcomes of the preregistered checkpoint rule—every unit ran its full 600-update budget and selection returned the unmoved initial checkpoint—not failed or abandoned fits. Arithmetic verified in both directions and against the per-seed sums. The study’s own gloss attributes the unmoved arms to $\beta \in \{0.1, 0.3\}$; three further arms select the unmoved channel at $\beta = 1.0$.

seed	arm	width	nominal rank loss	measured retained rank	post-erasure cross-cov. max
0	leace_dax16_none	16	10	6	$4.6e - 16$
0	leace_dax8_none	8	10	0	$1.9e - 35$
1	leace_dax16_none	16	10	6	$3.9e - 16$
1	leace_dax8_none	8	10	0	$3.2e - 35$
2	leace_dax16_none	16	10	7	$7.2e - 16$
2	leace_dax8_none	8	10	0	$4.0e - 35$

Table 11: Study 4 erasure rank, measured. The nominal rank loss is $1 + 8 + 1 = 10$ centred degrees of freedom for the joint $\{\text{SEX}, \text{RAC1P}, \text{public_coverage}\}$ schema. Counting does not determine the projection: at width 16 the *measured* loss is 10, 10, 9 across seeds, so seed 2 retains rank 7 where counting predicts 6. One RAC1P class carries a single complete case in the seed-0 fit fold (support [5989, 555, 90, 1, 32, 1674, 49, 1527, 596]). Rank-zero erasure at width 8 is established by measurement—retained rank 0 with post-erasure cross-covariance at 10^{-35} in all three seeds—not by the observation that $10 > 8$.

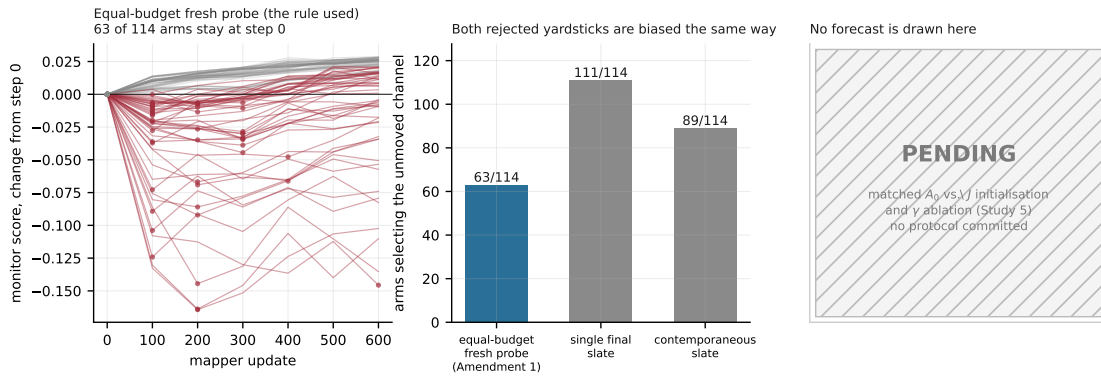


Figure 7: Study 4 monitor score along each of 114 trajectories relative to step 0; selection yardsticks; the matched initialisation and γ ablation shown empty here was run as Study 5 (§8).

I Study 5: full tables

track	nominal slots	fitted	distinct audited	exact duplicates
N (neural factorial)	168	168	102	66 (of untouched start)
E (projection)	204	204	192	12 (constant full-span maps)
total	372	372	294	78
references ref_A0 , ref_J (outside the 372)	6	–	6	–
stronger-attack interfaces (17 releases + <i>H</i> view, 3 anchors)	54	–	–	–
2017 transported Track E units (6 releases, 3 anchors)	18	–	–	–

Table 12: Study 5, from nominal slots to distinct audited releases. Duplicates are bitwise identical on every wire array of every pool and inherit the canonical audit; identity is never inferred from similar scores. The references, stress interfaces and transported units are separate counts, not subsets of the 372.

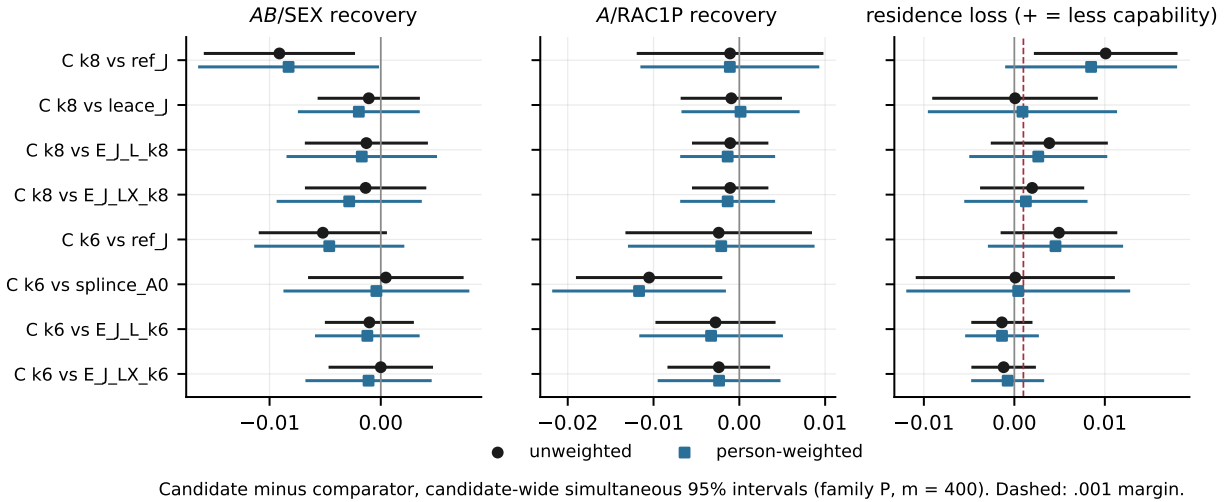


Figure 8: Study 5 panel contrasts, candidate minus comparator, candidate-wide simultaneous 95% intervals in family P ($m = 400$), both weightings. For recovery, negative is less disclosure; for residence loss, positive is less capability; dashed line: the 0.001 margin. Among the rows drawn, only the *J* comparison on *AB/SEX* (both weightings) and *SPLINCE* on *A/RAC1P* resolve in the candidate's favour; neither is accompanied by residence noninferiority. Table form with all endpoints: Appendix I.

release	comparator	fam.	AB/SEX	A/RAC1P	AB/RAC1P	residence loss
E_J_C_k8	ref_J	P	-0.0091 [-0.0158, -0.0025]	-0.0011 [-0.0118, +0.0096]	-0.0000 [-0.0054, +0.0053]	+0.0101 [+0.0023, +0.0179] [†]
E_J_C_k8	leace_J	P	-0.0011 [-0.0055, +0.0034]	-0.0009 [-0.0067, +0.0048]	-0.0000 [-0.0054, +0.0053]	+0.0001 [-0.0089, +0.0091]
E_J_C_k8	spline_A0	P	-0.0035 [-0.0101, +0.0032]	-0.0092 [-0.0165, -0.0018]	-0.0073 [-0.0150, +0.0004]	+0.0053 [-0.0045, +0.0150]
E_J_C_k8	E_J_L_k8	P	-0.0013 [-0.0067, +0.0041]	-0.0011 [-0.0053, +0.0032]	-0.0000 [-0.0054, +0.0053]	+0.0039 [-0.0024, +0.0102]
E_J_C_k8	E_J_LX_k8	P	-0.0014 [-0.0067, +0.0040]	-0.0011 [-0.0053, +0.0032]	-0.0000 [-0.0054, +0.0053]	+0.0020 [-0.0036, +0.0076]
E_J_C_k6	ref_J	P	-0.0052 [-0.0108, +0.0004]	-0.0024 [-0.0131, +0.0083]	-0.0017 [-0.0093, +0.0060]	+0.0049 [-0.0014, +0.0112]
E_J_C_k6	spline_A0	P	+0.0005 [-0.0064, +0.0073]	-0.0105 [-0.0189, -0.0022]	-0.0089 [-0.0178, -0.0001]	+0.0001 [-0.0107, +0.0109]
E_J_C_k6	E_J_L_k6	P	-0.0010 [-0.0049, +0.0028]	-0.0028 [-0.0096, +0.0040]	-0.0017 [-0.0093, +0.0060]	-0.0014 [-0.0046, +0.0018]
E_J_C_k6	E_J_LX_k6	P	+0.0000 [-0.0046, +0.0046]	-0.0024 [-0.0082, +0.0034]	-0.0017 [-0.0093, +0.0060]	-0.0012 [-0.0046, +0.0022]
E_A0_C_k2	E_A0_L_k2	X	+0.0051 [-0.0139, +0.0241]	-0.0202 [-0.0360, -0.0044]	-0.0253 [-0.0390, -0.0116]	-0.0018 [-0.0068, +0.0032]
E_A0_C_k2	E_A0_LX_k2	X	+0.0016 [-0.0127, +0.0158]	-0.0182 [-0.0328, -0.0035]	-0.0217 [-0.0351, -0.0082]	-0.0014 [-0.0062, +0.0035]

Table 13: Study 5 contrasts, unweighted, candidate minus comparator with candidate-wide simultaneous 95% intervals. Bold recovery: interval below zero. Bold residence with [†]: significantly worse. Rows marked X are in the exploratory whole-grid family ($m = 3900$) and were not nominated.

release	role	A/SEX	AB/SEX	A/RAC1P	AB/RAC1P	residence
A0	untouched weak channel	+0.0317	+0.0294	+0.0525	+0.0469	+0.0318
ref_J	strongest channel J (matched audit)	+0.0015	+0.0080	+0.0000	+0.0000	+0.0215
leace_A0	LEACE on A_0	+0.0039	+0.0038	+0.0097	+0.0096	+0.0220
splince_A0	SPLINCE on A_0	-0.0002	+0.0024	+0.0081	+0.0073	+0.0167
optnet16_C1	OptNet-ARL adaptation	+0.0070	+0.0046	+0.0060	+0.0116	+0.0170
leace_J	LEACE on J (diagnostic)	+0.0000	+0.0000	-0.0001	+0.0000	+0.0115
E_J_C_k8	panel: coalition projection, $k=8$	-0.0007	-0.0011	-0.0011	-0.0000	+0.0114
E_J_L_k8	local control, $k=8$	+0.0000	+0.0002	+0.0000	+0.0000	+0.0153
E_J_LX_k8	expanded local, $k=8$	+0.0006	+0.0003	+0.0000	+0.0000	+0.0134
E_J_C_k6	panel: coalition projection, $k=6$	-0.0004	+0.0028	-0.0024	-0.0017	+0.0166
E_J_L_k6	local control, $k=6$	-0.0008	+0.0039	+0.0004	+0.0000	+0.0152
E_J_LX_k6	expanded local, $k=6$	-0.0009	+0.0028	+0.0000	+0.0000	+0.0154
N_J_g000_C1_b100	panel: neural nominees (bitwise J)	+0.0015	+0.0080	+0.0000	+0.0000	+0.0215
E_A0_C_k2	exploratory: coalition on A_0 , $k=2$	+0.0163	+0.0201	+0.0223	+0.0183	+0.0309
E_A0_L_k2	local on A_0 , $k=2$	+0.0112	+0.0150	+0.0425	+0.0436	+0.0291
E_A0_LX_k2	expanded local on A_0 , $k=2$	+0.0114	+0.0185	+0.0404	+0.0400	+0.0296

Table 14: Study 5, 2018 development, unweighted seed means over three anchors: added recovery over H and residence gain over H . The neural nominees and every J -initialised arm at $\gamma \leq .01$ are bitwise J .

release	role	A/SEX	AB/SEX	A/RAC1P	AB/RAC1P	residence
A0	untouched weak channel	+0.0292	+0.0266	+0.0498	+0.0428	+0.0283
ref_J	strongest channel J (matched audit)	+0.0005	+0.0063	-0.0003	+0.0000	+0.0195
leace_A0	LEACE on A_0	+0.0027	+0.0033	+0.0097	+0.0094	+0.0205
splince_A0	SPLINCE on A_0	-0.0009	+0.0021	+0.0093	+0.0084	+0.0154
optnet16_C1	OptNet-ARL adaptation	+0.0067	+0.0039	+0.0038	+0.0094	+0.0169
leace_J	LEACE on J (diagnostic)	+0.0000	+0.0000	-0.0015	+0.0000	+0.0119
E_J_C_k8	panel: coalition projection, $k=8$	-0.0007	-0.0020	-0.0014	+0.0003	+0.0110
E_J_L_k8	local control, $k=8$	+0.0002	-0.0003	+0.0000	+0.0000	+0.0137
E_J_LX_k8	expanded local, $k=8$	+0.0009	+0.0009	+0.0000	+0.0000	+0.0123
E_J_C_k6	panel: coalition projection, $k=6$	-0.0004	+0.0017	-0.0024	-0.0036	+0.0150
E_J_L_k6	local control, $k=6$	-0.0010	+0.0029	+0.0009	+0.0000	+0.0136
E_J_LX_k6	expanded local, $k=6$	-0.0010	+0.0028	+0.0000	+0.0000	+0.0142
N_J_g000_C1_b100	panel: neural nominees (bitwise J)	+0.0005	+0.0063	-0.0003	+0.0000	+0.0195
E_A0_C_k2	exploratory: coalition on A_0 , $k=2$	+0.0145	+0.0170	+0.0224	+0.0170	+0.0295
E_A0_L_k2	local on A_0 , $k=2$	+0.0143	+0.0177	+0.0421	+0.0400	+0.0252
E_A0_LX_k2	expanded local on A_0 , $k=2$	+0.0122	+0.0184	+0.0401	+0.0384	+0.0261

Table 15: As above, person-weighted.

release	role	A/SEX	AB/SEX	A/RAC1P	AB/RAC1P	residence
A0	untouched weak channel	+0.0325	+0.0312	+0.0632	+0.0564	+0.0289
leace_A0	LEACE on A_0	+0.0071	+0.0072	+0.0145	+0.0101	+0.0219
splince_A0	SPLINCE on A_0	+0.0021	+0.0009	+0.0053	+0.0049	+0.0148
optnet16_C1	OptNet-ARL adaptation	+0.0093	+0.0088	+0.0211	+0.0185	+0.0139
E_J_C_k8	panel: coalition projection, $k=8$	-0.0010	+0.0006	-0.0009	+0.0008	+0.0116
E_J_L_k8	local control, $k=8$	-0.0001	+0.0009	-0.0012	-0.0002	+0.0131
E_J_LX_k8	expanded local, $k=8$	-0.0015	+0.0002	-0.0007	-0.0002	+0.0132
E_J_C_k6	panel: coalition projection, $k=6$	-0.0011	-0.0014	-0.0003	+0.0018	+0.0187
E_J_L_k6	local control, $k=6$	-0.0005	+0.0019	-0.0002	+0.0002	+0.0176
E_J_LX_k6	expanded local, $k=6$	-0.0010	-0.0002	-0.0006	+0.0011	+0.0172

Table 16: Study 5 exploratory 2017 transport, unweighted seed means; no intervals.

release	role	A/SEX	AB/SEX	A/RAC1P	AB/RAC1P	residence
A0	untouched weak channel	+0.0284	+0.0284	+0.0606	+0.0527	+0.0266
leace_A0	LEACE on A_0	+0.0048	+0.0065	+0.0127	+0.0084	+0.0197
splince_A0	SPLINCE on A_0	+0.0006	+0.0010	+0.0027	+0.0024	+0.0146
optnet16_C1	OptNet-ARL adaptation	+0.0077	+0.0080	+0.0205	+0.0179	+0.0135
E_J_C_k8	panel: coalition projection, $k=8$	-0.0026	-0.0011	-0.0019	+0.0006	+0.0105
E_J_L_k8	local control, $k=8$	-0.0012	-0.0003	-0.0013	-0.0007	+0.0117
E_J_LX_k8	expanded local, $k=8$	-0.0032	-0.0022	-0.0009	-0.0007	+0.0117
E_J_C_k6	panel: coalition projection, $k=6$	-0.0028	-0.0031	-0.0009	+0.0009	+0.0159
E_J_L_k6	local control, $k=6$	-0.0028	-0.0004	-0.0006	+0.0001	+0.0149
E_J_LX_k6	expanded local, $k=6$	-0.0042	-0.0035	-0.0007	+0.0006	+0.0149

Table 17: Study 5 exploratory 2017 transport, person-weighted seed means; no intervals.

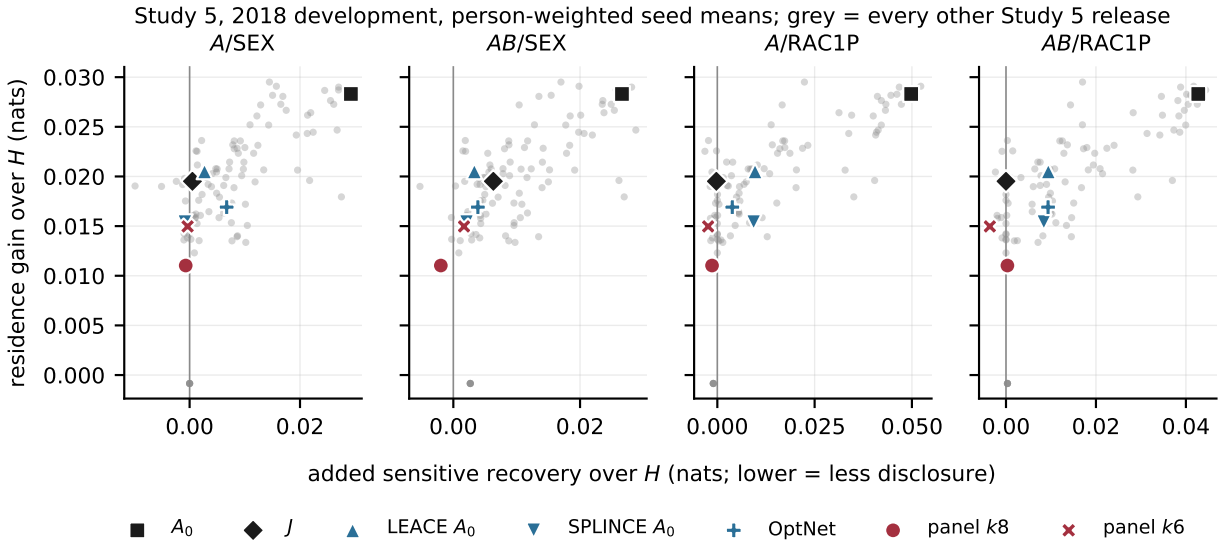


Figure 9: Study 5, 2018, person-weighted counterpart of Figure 4.

J Mathematical review of the Study 5 projection, as resolved

Form. With row vectors, $v = (z - \mu)R$ and $z_{\text{out}} = \mu + vP_kS$ compose to $z_{\text{out}} = \mu + (z - \mu)M$, $M = RP_kS$, idempotent and oblique (as is LEACE’s P^*). RS is the orthogonal projector onto the covariance support. Before the study we noted that under rank deficiency this whitened form deletes off-support components of new rows that LEACE’s subtractive form passes through, and that the form must be declared. The study declared the whitened form and measured supported rank 16 of 16 on both channels, every anchor, at every tolerance from 10^{-14} to 10^{-6} , in all 204 fits; the two forms therefore coincide, including on transported rows. Were a channel rank-deficient, the declared form would delete off-support directions; that case did not arise. The moment uses a globally centred v on declared masks.

Aliases and full span. $L2 = 2K_{\text{local}}$ is an exact eigenspace alias of L and was registered, not fitted. On these channels the moment Gram is full rank, so the full-span map is the constant μ , informationally equivalent to H (made exactly zero by amendment 2 so floating residue could not be amplified).

Stationarity, exact scope. For logits(h, z) = $\ell_{0,j}(h) + \Theta \cdot (b_j(h) \otimes v_{\text{out}})$ with $\ell_{0,j}$ fixed, average cross-entropy is convex in Θ with gradient $-G_j^{\text{out}}$ at zero; so $G_j^{\text{out}} = 0$ makes zero correction globally optimal *within that family on those rows*. Necessary conditions: the offset is the same cross-fitted $\ell_{0,j}$ that defines the

residual; the same masks and denominator; no free per-class intercept (its gradient $-\text{mean}(e_j)$ is not zeroed by cross-fitting, tested non-zero); linearity in $b \otimes v_{\text{out}}$. It says nothing about retrained H -only baselines, nonlinear auditors or population leakage; a tested counterexample has all linear moments near zero and S recovered by a quadratic probe. Partial projections generally leave non-zero training cross-moments (tested), so the statement does not apply to the partial nominees at all; it applies to full-span removal, which on these channels yields the constant map. **Nothing here equates removing linear dimensions with conditional privacy.**

Relation to earlier work. `acs_residual_spectral.py` builds per-role $\sum_c G_c G_c^\top = G G^\top$ matrices exactly equal to K_{local} , K_{coal} and the marginal kernel here, up to the input space and nuisance family; its release is the top eigenvectors of $U - \lambda P$, and as $\lambda \rightarrow \infty$ at retained rank $r - k$ these converge (off degeneracies) to the bottom $r - k$ eigenvectors of P . Study 5’s projection is that limit on a different input channel.

Claim	Status	Supporting measurement	What it does not establish
H preserved exactly in every release	Structural identity	Verified bitwise, 210/210 views	Says nothing about service <i>accuracy</i> , which moved by up to 0.015 nats
Appended channel adds authorised capability	Empirical, development & sealed	0.017–0.032 nats residence gain over H	Half-headroom reference fails in 1–3 of 3 seeds for every arm
Coalition penalty beats both local controls	Confirmatory (2017 sealed)	14/16 cells better, 0 worse, 2 unresolved	One year, one state, three seeds; residence cost not bounded below 0.001
Developed mechanism vs. frozen neural channel	Empirical, development	Worse on 3 of 4 sensitive endpoints	J is internal, and it supplies <i>less</i> residence capability: neither dominates
Rotation defect repaired exactly	Elementary algebra + numerics	Invariance to 10^{-15} , 18/18 cells	Invariance of an objective is not evidence the objective measures the right thing
Repair reduces measured disclosure	Refuted	8 of 48 cells significantly worse, 0 better	Attributable to the repaired <i>package</i> , not to any single changed ingredient
External adaptations beat the developed mechanism	Empirical, development	LEACE / SPLINCE / OptNet-C1 lower on all four endpoints	Adaptations, not replicas; LEACE costs residence against its own source channel
Any bound on $I(S; Z H)$	Not claimed	–	Finite validation-selected attacks attain no infimum in either direction
Study 5: competitive conjunction met by a panel release	Not established	0 of 4 nominees, both weightings, <code>ref_J</code> or historical J	Not a proof no competitive release exists; one objective, one selection rule
J -initialised training improves on J	Not supported	Selection returned J (step 0) in every J -arm at $\gamma \leq .01$	Other selection yardsticks untested
Coalition projection on J beats matched local projections	Not established	No candidate-wide difference vs L , LX (family P)	Nonsignificance is not equivalence
Projection of J lowers coalition sex recovery	Empirical, development	AB/SEX -0.0091 vs J , residence loss $+0.0101$ (unweighted)	A trade-off, not dominance; ordinary LEACE on J is as close as ± 0.0011 (points)
Coalition race effect on A_0 at $k=2$	Exploratory (family X)	Below both local controls, both weightings	Not nominated; residence unresolved, not preserved; far from J
Stronger attack confirms protection	Not claimed	MLP[256,256,128] weaker than standard audit	Survival untested
Study 6: extension adds capability on the proxy	Empirical, development	0.111–0.451 reduction in residual reconstruction error; 14/14 configurations	Reconstruction is a proxy, not residence transfer; no residential label entered it
Study 6: added disclosure within the declared .001-nat screen	Not established	0 of 14 configurations; closest $+0.0016$ unw / $+0.0021$ pw	Point-estimate screen, no intervals (nothing nominated); one family, $r=2$
Study 6: capability and disclosure separable by the adversarial term	Not established	Both fall monotonically in β in all three policies	This finite family and budget only
Study 6: coalition benefit over matched local control	Not established	$C1 + 0.0049$ vs $L2 + 0.0016$ at $\beta=30$	Third mechanism family to reproduce this
Utility-first candidate hidden in the earlier 372-slot evidence	None	0 of 124 configurations pass vs J or <code>leace_A0</code> ($m=2480$)	Exploratory reanalysis; does not change that study’s own verdict
Study 6: stronger attacker found	Not established	0 of 8 prespecified recipes beat the standard slate	Not evidence of survival against a stronger attack
Study 6 Tier 3 / Tier 4 results	NOT TRIGGERED	Gate rule after the failed pilot; no partial grid fitted	Absent evidence for an unrun design, not missing evidence for a claimed one

Table 18: Every load-bearing claim, its status, the measurement behind it, and what it does not establish. The traceable list—source commit, artifact, key, pool and uncertainty procedure—is `CLAIM_LEDGER.csv`.

K Study 6: protocol, counts and what was not run

tier	status	cost	outcome
T0 infrastructure, archive, restore	PASS	\$0.16	bytes restored and re-hashed; frozen mappers re-execute
T1 evidence reuse + audit calibration	PASS	\$0.20	0 of 124 stored configurations meet the utility-first criterion; 0 of 8 attacker recipes stronger than the standard slate
T2 pilot (42 units)	FAIL	\$0.35	capability leg passed for all 14 configurations; the .001-nat disclosure screen failed for all 14
T3 expansion (93 further slots)	NOT TRIGGERED	—	forbidden by the T2 gate rule; no partial grid was fitted
T4 nominee comparison, 2017 transport	NOT TRIGGERED	—	nothing was nominated

Table 19: Study 6 ran as a preregistered tier ladder with machine-readable gates. **Tiers 3 and 4 were never triggered:** the pilot gate failed and the protocol forbids opening a larger grid afterwards. They are absent evidence for an unrun design, not missing evidence for a claimed one; no partial Tier-3 grid was fitted. The pilot’s 42 units are the first 42 of the 135-slot cumulative plan.

Design. R has width $r = 2$; policies $L1$, $L2$ (mass-matched) and $C1$ (coalition) at $\beta \in \{1, 3, 10, 30\}$, plus two controls: an unprotected learned extension ($\beta = 0$) and a deployable PCA residual map. Differentiable attackers are corrections to a frozen service-plus- J baseline, so the incremental gain is zero at initialisation by construction; the extension decoder is zero-initialised, so the no-extension baseline is exactly recovered at step 0. **Counts.** 42 nominal pilot units (14 configurations $\times$ 3 anchors); 42 fitted; 42 audited (41 distinct, 1 exact duplicate); 9 reference audits (**ref_J**, **ref_A0**, **ref_leace_A0** $\times$ 3); 0 quarantines; 0 probability-integrity failures; 0 degenerate residual targets. 93 Tier-3 slots and every Tier-4 comparison were planned and not run: with the pilot’s 42, the cumulative expanded plan was 135 slots, and 42 of them exist. **Validation.** Frozen mappers re-execute from restored bytes; stored probes reproduce their predictions (log-loss difference 5.6×10^{-17}); H_A , H_B , Z_J byte-identical per unit and pool; subset-index hashes match the historical study; a routed-ancestor score check over 1,848 comparisons rejected none (maximum deviation 4.8×10^{-9} in log loss). The one failing check is cross-architecture audit reproduction (§10). **Amendment 1** (declared during Tier 1, before any pilot outcome) replaced an exact cross-architecture assertion with a tolerant one after an earlier Tier-1 attempt failed on it; the measurements and fixtures are committed with it. **Descriptive test-split frontier.** The study also reports a post-hoc test-split table that entered no gate and no selection; at $\beta = 30$ the protected arms sit within about .003 nats of J on residence and on the sensitive endpoints—which is what “the selection preferred nearly-untouched J ” looks like—while the unprotected control shows residence -0.0021 against $+0.027$ to $+0.035$ of added sensitive recovery. No claim rests on it.

L Corrections carried forward

From v4: Study 4 was never “a refinement of J ”; its $L2$ residence interval crosses 0.001; its coalition benefit is robust only against $L1$; unchanged channels were a selection outcome; the incumbents “found little more” only within their finite budget; rank-zero LEACE is a measurement, not a count; three of the 39 unmoved arms are at $\beta = 1$; the scope change is an evaluation correction. From v3: “LEACE dominates” is withdrawn; “indistinguishable from J ” is narrowed to unresolved; J is not Pareto-dominant; “implicit regulariser” is a hypothesis; the 63% rotation share is attained, not a supremum; a fitting-pool zero count is not population absence; the **n1r8-C1** deficit is $+0.01210$ person-weighted. The full texts are preserved at their manuscript commits (**ce5ba24a**, **72a63833**).

References

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing*

- Systems 36 (NeurIPS 2023)*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html. arXiv:2306.03819.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and causal parameters, 2016. URL <https://arxiv.org/abs/1608.00060>. journal version not verified in this audit.
- Sepehr Dehdashtian, Bashir Sadeghi, and Vishnu Naresh Boddeti. Utility-fairness trade-offs and how to find them. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. URL <https://arxiv.org/abs/2404.09454>. arXiv:2404.09454.
- Frances Ding, Moritz Hardt, John P. Miller, and Ludwig Schmidt. Retiring adult: New datasets for fair machine learning. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21, Brussels, Belgium, 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1002. URL <https://aclanthology.org/D18-1002/>.
- Ky Fan. On a theorem of Weyl concerning eigenvalues of linear transformations I. *Proceedings of the National Academy of Sciences of the United States of America*, 35(11):652–655, 1949. doi: 10.1073/pnas.35.11.652.
- Floris Holstege, Shauli Ravfogel, and Bram Wouters. Preserving task-relevant information under linear concept removal. In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025. URL <https://arxiv.org/abs/2506.10703>. arXiv:2506.10703.
- David Madras, Elliot Creager, Toniann Pitassi, and Richard Zemel. Learning adversarially fair and transferable representations. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3384–3393. PMLR, 2018. URL <https://proceedings.mlr.press/v80/madras18a.html>.
- David McAllester and Karl Stratos. Formal limitations on the measurement of mutual information. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 875–884. PMLR, 2020. URL <https://proceedings.mlr.press/v108/mcallester20a.html>. arXiv:1811.04251.
- Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems 20 (NIPS 2007)*, 2007. URL https://papers.nips.cc/paper_files/paper/2007/hash/013a006f03dbc5392effeb8f18fda755-Abstract.html.
- Bashir Sadeghi, Runyi Yu, and Vishnu Naresh Boddeti. On the global optima of kernelized adversarial representation learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. URL <https://arxiv.org/abs/1910.07423>. arXiv:1910.07423.
- Bashir Sadeghi, Lan Wang, and Vishnu Naresh Boddeti. Adversarial representation learning with closed-form solvers. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2021)*, pages 731–748, 2021. doi: 10.1007/978-3-030-86520-7_45. URL <https://arxiv.org/abs/2109.05535>. arXiv:2109.05535; pages and DOI from dblp/search listing (Springer page not loadable).
- Bashir Sadeghi, Sepehr Dehdashtian, and Vishnu Naresh Boddeti. On characterizing the trade-off in invariant representation learning. *Transactions on Machine Learning Research*, 2022. URL <https://arxiv.org/abs/2109.03386>. arXiv:2109.03386; OpenReview id 3gfpBR1ncr.
- Lalitha Sankar, S. Raj Rajagopalan, and H. Vincent Poor. Utility-privacy tradeoffs in databases: An information-theoretic approach. *IEEE Transactions on Information Forensics and Security*, 2013. doi: 10.1109/TIFS.2013.2253320. URL <https://arxiv.org/abs/1102.3751>. arXiv:1102.3751.

- Theresa Stadler, Bogdan Kulynych, Michael Gastpar, Nicolas Papernot, and Carmela Troncoso. The fundamental limits of least-privilege learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 46393–46411. PMLR, 2024. URL <https://proceedings.mlr.press/v235/stadler24a.html>.
- Eric V. Strobl, Kun Zhang, and Shyam Visweswaran. Approximate kernel-based conditional independence tests for fast non-parametric causal discovery. *Journal of Causal Inference*, 7(1), 2019. doi: 10.1515/jci-2018-0017. URL <https://arxiv.org/abs/1702.03877>. arXiv:1702.03877.
- Kun Zhang, Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. Kernel-based conditional independence test and application in causal discovery. In *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 804–813, 2011. URL <https://arxiv.org/abs/1202.3775>. arXiv:1202.3775.